Exercise J-M

1. Let $y = y(x)$ be the solution of the differential equation $e^x\sqrt{1-y^2}\,dx + \left(\frac{y}{x}\right)dy = 0, y(1) = -1.$ Then the value of $(y(3))^2$ is equal to : **[JEE(Main)July2021]**

माना अवकल समीकरण $e^x\sqrt{1-y^2}\,dx + \left(\frac{y}{x}\right)dy = 0, y(1) = -1$ का हल $y = y(x)$ है। तो $(y(3))^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $1+4e^3$ (B) $1-4e^3$ (C) $1+4e^6$ (D) $1-4e^6$

Ans. D

Sol. $e^x\sqrt{1-y^2}\,dx + \left(\frac{y}{x}\right)dy = 0$

$$\int xe^x dx + \int \frac{y}{\sqrt{1-y^2}}dy = 0$$

$$e^x(x-1) - \sqrt{1-y^2} = c$$

Put $x = 1, \quad y = 1 \Rightarrow c = 0$

$$e^x(x-1) = \sqrt{1-y^2}$$

Put $x = 3$

$$e^3(3-1) = \sqrt{1-y^2}$$

$$2e^3 = \sqrt{1-y^2}$$

$4e^6 = 1 - y^2$

$y^2 = 1 - 4e^6$

$\Rightarrow (y(3))^2 = 1 - 4e^6$

2. Let $y = y(x)$ satisfies the equation $\frac{dy}{dx} - |A| = 0,$ for all $x > 0$, where $A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}$. If $y(\pi) = \pi + 2$, then the value of $y\left(\frac{\pi}{2}\right)$ is : **[JEE(Main)July2021]**

माना $y = y(x)$, समीकरण $\frac{dy}{dx} - |A| = 0,\ \forall\ x > 0$, को संतुष्ट करता है, जबकि $A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}$ है। यदि $y(\pi) = \pi + 2$ है, तो $y\left(\frac{\pi}{2}\right)$ बराबर है : **[JEE(Main)July2021]**

(A) $\frac{\pi}{2} + \frac{4}{\pi}$ (B) $\frac{3\pi}{2} - \frac{1}{\pi}$ (C) $\frac{\pi}{2} - \frac{4}{\pi}$ (D) $\frac{\pi}{2} - \frac{1}{\pi}$

Ans. A

Sol. $\frac{dy}{dx} = |A| = \begin{vmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{vmatrix}$

expand along R_3

$\frac{dy}{dx} = 2(\sin x + 1) + \frac{1}{x}(-y)$

$\frac{dy}{dx} = 2(\sin x + 1) - \frac{y}{x}$

$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = 2(\sin x + 1)$

$I.F. = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$

$x\frac{dy}{dx} + y = 2x(\sin x + 1)$

$d(xy) = \int 2x(\sin x + 1)dx$

$\frac{xy}{2} = -x\cos x - \int(-\cos x)dx + \frac{x^2}{2} + C$

$\frac{xy}{2} = -x\cos x + \sin x + \frac{x^2}{2} + C$

$\frac{\pi(\pi+2)}{2} = -\pi(-1) + 0 + \frac{\pi^2}{2} + C$

$C = 0$

$$\frac{\frac{\pi}{2}y\left(\frac{\pi}{2}\right)}{2} = 0 + 1 + \frac{\pi^2}{8} \quad \Rightarrow y$$

$$y\left(\frac{\pi}{2}\right) = \frac{4}{\pi} + \frac{\pi}{2}$$

3. Let y = y (x) be the solution of the differential equation $\text{cosec}^2 x\ dy + 2dx = (1 + y\cos 2x)\ \text{cosec}^2 x\ dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0) + 1)^2$ is equal to : **[JEE(Main)July2021]**

माना अवकल समीकरण $\text{cosec}^2 x\ dy + 2dx = (1 + y\cos 2x)\ \text{cosec}^2 x\ dx$, जबकि $y\left(\frac{\pi}{4}\right) = 0$ है, का हल y = y (x) है। तो $(y(0) + 1)^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) e^{-1} (B) e (C) $e^{\frac{-1}{2}}$ (D) $e^{\frac{1}{2}}$

Ans. A

Sol. $\text{cosec}^2 x\, dy + 2dx = (1 + y\cos 2x)\ \text{cosec}^2 x$

$$\frac{dy}{dx} + 2\sin^2 x = 1 + y\cos 2x$$

$$\frac{dy}{dx} + y(-\cos 2x) = 1 - 2\sin^2 x$$

$$\frac{dy}{dx} + y(-\cos 2x) = \cos 2x$$

$$\frac{dy}{dx} + Py = Q$$

$$\text{I.F.} = e^{\int -\cos 2x\, dx}$$

$$= e^{-\frac{1}{2}\sin 2x}$$

$$y(\text{I.F.}) = \int (\text{I.F.})\, Q dx$$

$$y\left(e^{-\frac{1}{2}\sin 2x}\right) = \int e^{-\frac{1}{2}\sin 2x} \cos 2x\, dx$$

$$-\frac{1}{2}\sin 2x = t$$

$$-\cos 2x\,dx = dt$$

$$\cos 2x\,dx = -dt$$

$$y\left(e^{-\frac{1}{2}\sin 2x}\right) = \int e^{t}(-dt)$$

$$ye^{-\frac{1}{2}\sin 2x} = -e^{t} + C$$

$$ye^{-\frac{1}{2}\sin 2x} = -e^{-\frac{1}{2}\sin 2x} + C$$

by $y\left(\frac{\pi}{4}\right) = 0$

$$0 = -e^{-\frac{1}{2}} + C$$

$$C = \frac{1}{\sqrt{e}}$$

$$ye^{-\frac{1}{2}\sin 2x} = -e^{-\frac{1}{2}\sin 2x} + \frac{1}{\sqrt{C}}$$

at $x = 0$

$$y(0) = -1 + \frac{1}{\sqrt{e}}$$

$$\left(y(0)+1\right)^2 = \left(-1 + \frac{1}{\sqrt{e}} + 1\right)^2 = \left(e^{-\frac{1}{2}}\right)^2$$

$$= e^{-1}$$

4. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 1 + x\,e^{y-x}$, $-\sqrt{2} < x < \sqrt{2}$, $y(0) = 0$ then, the minimum value of $y(x)$, $x \in \left(-\sqrt{2}, \sqrt{2}\right)$ is equal to : **[JEE(Main)July2021]**

माना अवकल समीकरण, $\frac{dy}{dx} = 1 + x\,e^{y-x}$, $-\sqrt{2} < x < \sqrt{2}$, $y(0) = 0$ का हल $y = y(x)$ है। तो $y(x)$, $x \in \left(-\sqrt{2}, \sqrt{2}\right)$ में $y(x)$ का न्यूनतम मान बराबर है : **[JEE(Main)July2021]**

(A) $\left(2-\sqrt{3}\right) - \log_e 2$

(B) $\left(1-\sqrt{3}\right) - \log_e\left(\sqrt{3}-1\right)$

(C) $\left(2+\sqrt{3}\right) + \log_e 2$

(D) $\left(1+\sqrt{3}\right) - \log_e\left(\sqrt{3}-1\right)$

Ans. B

Sol. $e^{-y}.\dfrac{dy}{dx}-e^{-y}=xe^{-x}$

$-e^{-y}=t$

$e^{-y}\dfrac{dy}{dx}=\dfrac{dt}{dx}$

$\dfrac{dt}{dx}+t=xe^{-x}$

$IF=\int\limits_{e}dx=e^{x}$

$t.e^{x}=\int x.e^{-x}.e^{x}dx$

$t.e^{x}=\dfrac{x^2}{2}+C$

$-e^{-y}e^{x}=\dfrac{x^2}{2}+C$

$\because y(0)=0$

$-(1)(1)=0+c\Rightarrow c=-1$

$-e^{-y}e^{x}=\dfrac{x^2}{2}-1$

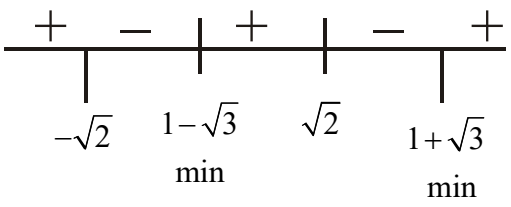
$e^{x-y}=\dfrac{2-x^2}{2}$

$x-y=\ln\left(\dfrac{2-x^2}{2}\right)$

$y=x-\ln\left(\dfrac{2-x^2}{2}\right)$

$\dfrac{dy}{dx}=1-\dfrac{2}{2-x^2}\left(-\dfrac{2x}{2}\right)=1+\dfrac{2x}{2-x^2}=\dfrac{x^2-2-2x}{x^2-2}$

$\dfrac{dy}{dx}=\dfrac{(x-(1+\sqrt{3}))(x-(1-\sqrt{3}))}{(x-\sqrt{2})(x+\sqrt{2})}$



$\because x \in (-\sqrt{2}, \sqrt{2})$

minima at $x = 1-\sqrt{3}$

$y = 1-\sqrt{3} - \ln(\sqrt{3}-1)$

5. Let $y = y(x)$ be the solution of the differential equation $xdy = (y + x^3\cos x)\,dx$ with $y(\pi) = 0$, then $y\left(\frac{\pi}{2}\right)$ is equla to : **[JEE(Main)July2021]**

माना अवकल समीकरण $xdy = (y + x^3\cos x)dx$, $y(\pi) = 0$, का हल $y = y(x)$ है। तो $y\left(\frac{\pi}{2}\right)$ बराबर है : **[JEE(Main)July2021]**

(A) $\frac{\pi^2}{2}-\frac{\pi}{4}$ (B) $\frac{\pi^2}{4}-\frac{\pi}{2}$ (C) $\frac{\pi^2}{4}+\frac{\pi}{2}$ (D) $\frac{\pi^2}{2}+\frac{\pi}{4}$

Ans. C

Sol. $xdy = ydx + x^3\cos x\,dy$

$$\frac{xdy - ydx}{x^2} = x\cos x\,dx$$

$$\int d(y/x) = \int x\cos x\ dx$$

$$\frac{y}{x} = x\int \cos x\ dx - \int\left(dx\int \cos x\,dx\right)dx$$

$$\frac{y}{x} = x\sin x - \int \sin x\ dx$$

$$\frac{y}{x} = x\sin x + \cos x + C$$

$$\frac{0}{\pi} = \pi\sin\pi + \cos\pi + C$$

$0 = 0 - 1 + C$

$C = 1$

$$\frac{y}{x} = x\sin x + \cos x + 1$$

$$\frac{y\times 2}{\pi} = \frac{\pi}{2}\sin\frac{\pi}{2} + \cos\frac{\pi}{2} + 1$$

$$y\times\frac{2}{\pi} = \frac{\pi}{2} + 1$$

$$y = \frac{\pi^2}{4} + \frac{\pi}{2}$$

6. Let $y = y(x)$ be solution of the differentiable equation $\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with $y(0) = 0$. If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha\log_e 2$, then the value of α is equal to : **[JEE(Main)July2021]**

माना अवकल समीकरण $\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, $y(0) = 0$ का हल $y = y(x)$ है। यदि $y\left(-\frac{2}{3}\log_e 2\right) = \alpha \log_e 2$ है, तो α का मान बराबर है : **[JEE(Main)July2021]**

(A) 2 (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{2}$

Ans. B

Sol. $\frac{dy}{dx} = e^{3x+4y}$

$\Rightarrow \int e^{-4y}dy = \int e^{3x}\,dx$

$\Rightarrow \frac{-e^{-4y}}{4} = \frac{e^{3x}}{3} + C \qquad (y(0) = 0)$

$\frac{-1}{4} = \frac{1}{3} + C \Rightarrow C = \frac{-7}{12}$

$\Rightarrow \frac{-e^{4y}}{4} = \frac{e^{3x}}{3} - \frac{7}{12}$

$\Rightarrow -3e^{-4y} = 4e^{3x} - 7$

$\Rightarrow e^{-4y} = \frac{7 - 4e^{3x}}{3}$

$y = \frac{-1}{4}\,ln\left(\frac{7 - 4e^{3x}}{3}\right)$

$y\left(\frac{-2}{3}\log_e 2\right) = \frac{1}{4}\,ln\left(\frac{7 - 4e^{3\left(\frac{-2}{3}\log_e 2\right)}}{3}\right)$

$= \frac{-1}{4}\,ln\,2$

7. Let $y = y(x)$ be the solution of the differential equation $(x - x^3)dy = (y + yx^2 - 3x^4)dx$, $x > 2$. If $y(3) = 3$, then $y(4)$ is equal to : **[JEE(Main)July2021]**

माना अवकल समीकरण $(x - x^3)dy = (y + yx^2 - 3x^4)dx$, $x > 2$ का हल $y = y(x)$ है। यदि $y = 3$ हैं, तो y बराबर है : **[JEE(Main)July2021]**

(A) 12 (B) 16 (C) 8 (D) 4

Ans. A

Sol. $(x - x^3)dy = (y + yx^2 - 3x^4)\,dx$

$\Rightarrow x\,dy - y\,dx = x^2(xdy + ydx) - 3x^4\,dx$

$\Rightarrow = \dfrac{xdy - ydx}{x^2} = xdy + ydx - 3x^2dx$

$\Rightarrow \left(\dfrac{y}{x}\right) = d(xy) - d(x^3)$

Integrating both sides

$\dfrac{y}{x} = xy - x^3 + c$

$\therefore\ y(3) = 3$

$\dfrac{3}{3} = 3 \times 3 - 27 + c$

$c = 19$

$\therefore\ \dfrac{y}{x} = xy - x^3 + 19$ . .....(1)

for $y(4)$, substitute $x = 4$ in .....(1)

$\dfrac{y(4)}{4} = 4y(4) - 64 + 19$

$\Rightarrow\ y(4) = 16y(4) - 4 \times 45$

$15y(4) = 4 \times 55$

$y(4) = \dfrac{4 \times 45}{15} = 12$

8. Let a curve $y = y(x)$ be given by the solution of the differential equation $\cos\left(\dfrac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x} - 1}\,dy$. If it intersects y-axis at $y = -1$, and the intersection point of the curve with x-axis is $(\alpha, 0)$, then e^{α} is equal to _____. **[JEE(Main)July2021]**

माना वक्र $y = y(x)$, अवकल समीकरण $\cos\left(\dfrac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x} - 1}\,dy$ का हल है। यदि वह y-अक्ष को $y = -1$ तथा x-अक्ष को $(\alpha, 0)$ पर काटता है, तो e^{α} बराबर है _____ **[JEE(Main)July2021]**

Ans. 2

Sol. $\cos\left(\frac{1}{2}\cos^{-1}\left(e^{-x}\right)\right)dx = \sqrt{e^{2x}-1}\ dy$

Let $\cos^{-1} e^{-x} = \theta$

$\cos\theta = e^{-x}$

$$\cos\frac{\theta}{2} = \sqrt{\frac{e^{-x}+1}{2}}$$

$$\sqrt{\frac{e^{-x}+1}{2}}\ dx = \sqrt{e^{x}+1}\ \sqrt{e^{x}-1}\ dy$$

$$\sqrt{\frac{1+e^{x}}{2e^{x}}}\ dx = \sqrt{e^{x}+1}\ \sqrt{e^{x}-1}\ dy$$

$$\int \frac{dx}{\sqrt{2e^{x}}\sqrt{e^{x}-1}} = \int dy$$

$$\frac{1}{\sqrt{2}}\int \frac{e^{-x}dx}{1-e^{-x}} = y + c$$

$1 - e^{-x} = t^2$

$e^{-x}\,dx = 2t\,dx$

$$\frac{1}{\sqrt{2}}\int \frac{2t\ dt}{t} = y + c$$

$$\sqrt{2}\sqrt{1-e^{-x}} = y + c$$

$$\sqrt{2}\sqrt{1-e^{-x}} = y + 1$$

$y = 0$

$e^x = 2$

$c = 1$

9. Let y = y (x) be solution of the differential equation $\left(\left(x+2\right)e^{\left(\frac{y+1}{x+2}\right)}+\left(y+1\right)\right)dx = (x+2)dy$, y(1) = 1. If the domain of y = y(x) is an open interval (α,β), then $|\alpha+\beta|$ is equal to. **[JEE(Main)July2021]**

माना अवकल समीकरण $\left(\left(x+2\right)e^{\left(\frac{y+1}{x+2}\right)}+\left(y+1\right)\right)dx = (x+2)dy$, y(1) = 1 का हल, y = y(x) है। यदि y = y(x) का प्रान्त विवृत अन्तराल (α, β) हैं, तो $|\alpha + \beta|$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 4

Sol. $\left((x+2)e^{\frac{y+1}{x+2}}+(y+1)\right)dx=(x+2)dy$

$x+2=X \quad \Rightarrow \quad dx=dX$

$y+1=Y \quad \Rightarrow \quad dy=dY$

$\left(Xe^{\frac{Y}{X}}+Y\right)dX=X\,dY$

$\frac{dY}{dX}=e^{\frac{Y}{X}}+\frac{Y}{X}$

put $Y=tX$

$\frac{dY}{dx}=t+X\frac{dt}{dX}$

$\Rightarrow \quad t+X\frac{dt}{dX}=e^{t}+t$

$\Rightarrow \quad e^{-t}dt=\frac{dX}{X}$

$\Rightarrow \quad -e^{-t}=\ln|X|+C$

$\Rightarrow \quad -e^{-\frac{Y}{X}}=\ln|X|+C$

Now this pass through (3, 2)

$-e^{-\frac{2}{3}}=\ln 3+C$

then

$-e^{-\frac{Y}{X}}=\ln|X|-e^{-\frac{2}{3}}-\ln 3$

$-e^{-\frac{Y}{X}}=\ln 3+e^{-\frac{2}{3}}-\ln|X|>0$

$\ln|X|<e^{-\frac{2}{3}}+\ln 3$

let $\lambda=e^{-\frac{2}{3}}+\ln 3$

$\ln|X|<\lambda$

$|X|<e^{\lambda}$

$|x+2|<e^{\lambda}$

$-e^{\lambda} < x+2 < e^{\lambda}$

$-e^{\lambda} - 2 < x < e^{\lambda} - 2$

$\alpha = -e^{\lambda} - 2$

$\beta = e^{\lambda} - 2$

$|\alpha + \beta| = 4$

10. Let $y = y(x)$ be solution of the follwoing differential equation $e^{y}\frac{dy}{dx} - 2e^{y}\sin x + \sin x\cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$. If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha+\beta)$ is equal to _____. **[JEE(Main)July2021]**

माना $y = y(x)$, अवकल समीकरण $e^{y}\frac{dy}{dx} - 2e^{y}\sin x + \sin x\cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$ का हल है। यदि $y(0) = \log_e(\alpha + \beta e^{-2})$ है, तो $4(\alpha+\beta)$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 4

Sol. $e^{y}\frac{dy}{dx} - 2e^{y}\sin x = -\sin x\cos^2 x$

Put $e^{y} = t$

$e^{y}\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} - 2t\sin x = -\sin x\cos^2 x$

I.F. $= e^{-\int 2\sin x dx} = e^{2\cos x}$

$e^{y}.e^{2\cos x} = \int e^{2\cos x}(-\sin x_\cos^2 x dx$

$\cos x = z$

$= \int p^{2z}z^2 dz$

$\frac{e^{2z}}{2}z^2 - \int e^{2z}.z dz$

$= \frac{e^{2z}}{2}z^2 - \int e^{2z}.z dz$

$= \frac{e^{2z}}{2}z^2 - \left[\frac{e^{2z}}{2}.z - \frac{e^{2z}}{4}\right] + c$

$\Rightarrow e^{y}e^{2\cos x} = \frac{e^{2z}}{4}(2z^2 - 2z + 1) + c$

$$\Rightarrow e^y e^{2\cos x} = \frac{e^{2\cos x}}{4}(2\cos^2 x - 2\cos x + 1) + c$$

$$\Rightarrow e^y = \frac{1}{4}(2\cos^2 x - 2\cos x + 1) + c$$

At $x = \frac{\pi}{2} y = 0,$

$$\Rightarrow 1 = \frac{1}{4} + c \Rightarrow c = \frac{3}{4}$$

$$\Rightarrow e^y = \frac{1}{4}(2\cos^2 x - 2\cos x + 1) + \frac{3}{4}e^{-2\cos x}$$

$$\Rightarrow y(0) = \ell n\left(\frac{1}{4} + \frac{3}{4}e^{-2}\right)$$

$$\Rightarrow y(0) = \ell n(\alpha + \beta e^{-2})$$

$$\Rightarrow \alpha = \frac{1}{4}, \beta = \frac{3}{4}$$

$$\Rightarrow 4(\alpha + \beta) = 4$$

11. Let a curve $y = f(x)$ pass through the point $(2, (\log_e 2)^2)$ and have slope $\frac{2y}{x \log_e x}$ for all positive real value of x. Then the value of $f(e)$ is equal to _____. **[JEE(Main)July2021]**

माना एक वक्र $y = f(x)$, बिन्दु $(2, (\log_e 2)^2)$ से होकर जाता है तथा x के सभी धनात्मक वास्तविक मानों के लिए इसकी प्रवणता $\frac{2y}{x \log_e x}$ है, तो $f(e)$ का मान है _____ **[JEE(Main)July2021]**

Ans. 1

Sol. $$\frac{dy}{dx} = \frac{2y}{x \ln x}$$

$$\frac{dy}{2y} = \frac{dx}{x \ln x}$$

$$\frac{1}{2}\int \frac{dy}{y} = \int \frac{dx}{x \ln x}$$

$$\frac{1}{2}\ln y = \int \frac{dx}{x \ln x}$$

$$\ln x = t$$

$$\frac{dx}{x} = dt$$

$$\frac{\ln y}{2} = \int \frac{dt}{t} \Rightarrow \frac{\ln y}{2} = \ln t + c$$

$$\frac{\ln y}{2} = \ln(\ln x) + c$$

$$\frac{\ln(\ln 2)^2}{2} = \ln(\ln 2) + c$$

$$\frac{2\ln(\ln 2)}{2} = \ln(\ln 2) + c$$

$C = 0$

$$\frac{\ln y}{2} = \ln(\ln x)$$

$$y = e^{2\ln(\ln x)}$$

$$f(x) = e^{2\ln(\ln x)}$$

$f(e) = e^{2\ln(\ln e)} = 1$

12. If $y = y(x)$, $y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation $\sec y \frac{dy}{dx} - \sin(x + y) - \sin(x - y) = 0$, with $y(0) = 0$, then $5y'\left(\frac{\pi}{2}\right)$ is equal to_____. **[JEE(Main)July2021]**

यदि अवकल समीकरण $\sec y \frac{dy}{dx} - \sin(x + y) - \sin(x - y) = 0$, $y(0) = 0$ का हल $y = y(x)$, $y \in \left[0, \frac{\pi}{2}\right)$ है, तो $5y'\left(\frac{\pi}{2}\right)$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 2

Sol. $\sec y \frac{dy}{dx} = 2 \sin x \cos y$

$\Rightarrow \int \sec^2 y \, dy = \int 2 \sin x \, dx$

$\Rightarrow \tan y = -2 \cos x + C$

$y(0) = 0 \Rightarrow C = 2$

$\tan y = 2 - 2 \cos x$

at $x = \frac{\pi}{2}$ $\tan y = 2$

$\sec^2 y \frac{dy}{dx} = 2 \sin x \Rightarrow \frac{dy}{dx} = \frac{2 \sin x}{1 + \tan^2 y}$

$\left.\frac{dy}{dx}\right|_{x=\frac{\pi}{2}} = \frac{2}{1+4} = \frac{2}{5}$

$5y'\left(\frac{\pi}{2}\right) = 2$

13. Let y = y(x) be the solution of the differential equation $dy = e^{\alpha x + y} dx$; $\alpha \in N$. If $y(\log_e 2) = \log_e 2$ and $y(0) = \log_e \left(\frac{1}{2}\right)$, then the value of α is equal to _____. **[JEE(Main)July2021]**

माना अवकल समीकरण $dy = e^{\alpha x + y} dx$; $\alpha \in N$ का हल $y = y(x)$ है। यदि $y(\log_e 2) = \log_e 2$ तथा $y(0) = \log_e \left(\frac{1}{2}\right)$ है, तो α का मान बराबर है _____ **[JEE(Main)July2021]**

Ans. 2

Sol. $dy = e^{\alpha x + y} dx$

$\Rightarrow e^{-y} dy = e^{\alpha x} dx$

Integrating both sides.

$-e^{-y} = \frac{e^{\alpha x}}{\alpha} + C$ .....(i)

$\therefore y(\ln 2) = \ln 2$

$\therefore -e^{-\ln 2} = \frac{e^{\alpha \ln 2}}{\alpha} + C$

$\Rightarrow -\frac{1}{2} = \frac{2^{\alpha}}{\alpha} + C$ .....(ii)

$\therefore y(0) = \ln\left(\frac{1}{2}\right)$

$\Rightarrow -e^{-\ln\left(\frac{1}{2}\right)} = \frac{1}{\alpha} + C$

$\Rightarrow -2 = \frac{1}{\alpha} + C$ .....(iii)

(ii)-(iii)

$\therefore\ \alpha \in N$

$\frac{3}{2}=\frac{2^{\alpha}-1}{\alpha}$

$\alpha = 1$, $\frac{3}{2}=\frac{2-1}{1}$ (False)

$\alpha = 2$, $\frac{3}{2}=\frac{2^2-1}{1}$ (True)

$\therefore$ Value of $\alpha = 2$

14. Let $y = y(x)$ be a solution curve of the differential equation $(y+1)\tan^2 x\,dx + \tan x\,dy + ydx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$.

If $\lim\limits_{x\to 0+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is : **[JEE(Main)Aug.2021]**

माना अवकल समीकरण $(y+1)\tan^2 x\,dx + \tan x\,dy + ydx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$ का हल $y = y(x)$ है। यदि $\lim\limits_{x\to 0+} xy(x) = 1$ है,

तो $y\left(\frac{\pi}{4}\right)$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{4}+1$ (D) $\frac{\pi}{4}-1$

Ans. A

Sol. $\tan x\frac{dy}{dx} + y + (y+1)\tan^2 x = 0$

$\tan x\frac{dy}{dx} + y\sec^2 x = -\tan^2 x$

$\int d(y\tan x) = -\int \tan^2 x\,dx$

$y\tan x = x - \tan x + C$

$y = x\cot x - 1 + C\cot x$

$\lim\limits_{x\to 0^+} xy = \lim\limits_{x\to 0}(x^2\cot x - x + xC\cot x) = 1$

$y = x\cot x - 1 + \cot x$

Put $x = \frac{\pi}{4}$

$$y\left(\frac{\pi}{4}\right)=\frac{\pi}{4}-1+1=\frac{\pi}{4}$$

15. Let y(x) be the solution of the differential equation $2x^2\,dy+(e^y-2x)\,dx=0$, $x>0$. If $y(e)=1$, then $y(1)$ is equal to : **[JEE(Main)Aug.2021]**

माना अवकल समीकरण $2x^2\,dy+(e^y-2x)\,dx=0$, $x>0$ का हल y(x) है। यदि $y(e)=1$ है, तो $y(1)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 2 (B) $\log_e(2e)$ (C) 0 (D) $\log_e 2$

Ans. D

Sol. $\frac{dy}{dx}-\frac{-e^y}{2x^2}+\frac{1}{x}$

$$\frac{e^{-y}dy}{dx}=\frac{e^y}{x}+\frac{-1}{2x^2}$$

Let $e^{-y}=t$

$$\frac{dt}{dx}+\frac{t}{x}=\frac{1}{2x^2}\quad\left\{\text{I.F.}=e^{\int\frac{1}{x}dx}=x\right\}$$

$$tx=\int\frac{1}{2x}dx$$

$$e^{-y}x=\frac{1}{2}\ln x+c$$

$$c=\frac{1}{2}$$

$$e_x^{-y}=\frac{1}{2}(1+\ln x)$$

16. Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}=2\,(y+2\sin x-5)\,x-2\cos x$ such that $y(0)=7$. Then $y(\pi)$ is equal to : **[JEE(Main)Aug.2021]**

माना अवकल समीकरण $\frac{dy}{dx}=2\,(y+2\sin x-5)\,x-2\cos x$ का हल $y=y(x)$ है, जिसके लिए $y(0)=7$ है तो $y(\pi)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $7e^{\pi^2}+5$ (B) $2e^{\pi^2}+5$ (C) $3e^{\pi^2}+5$ (D) $e^{\pi^2}+5$

Ans. B

Sol. $\frac{dy}{dx} - 2xy = 2x(2\sin x - 5) - 2\cos x$

I.F. $= e^{\int -2x dx} = e^{-x^2}$

$$y \cdot e^{-x^2} = \int e^{-x^2} \left\{2x\left(2\sin x - 5\right) - 2\cos x\right\} dt$$

$$y \cdot e^{-x^2} = \int e^{-x^2} \left\{(-2x)(-2\sin x) + (-2\cos x)\right\} dx + 5\int e^{-x^2}(-2x)dx$$

$$y \cdot e^{-x^2} = e^{-x^2}\left(-2\sin x\right) + 5e^{-x^2} + C$$

$$y = 5 - 2\sin x + C \cdot e^{x^2}$$

$$y(0) = 7 \Rightarrow C = 2$$

$$y = 5 - 2\sin x + 2 \cdot e^{x^2}$$

$$y(\pi) = 5 + 2e^{\pi^2}$$

17. Let us consider a curve, y = f(x) passing through the point (–2, 2) and the slope of the tangent to the curve at any point (x, f(x)) is given by $f(x) + xf'(x) = x^2$. Then : **[JEE(Main)Aug.2021]**

माना एक वक्र y = f(x) बिन्दु (–2, 2) से होकर जाता है तथा वक्र के किसी बिन्दु (x, f(x)) पर स्पर्शरेखा की प्रवणता $f(x) + xf'(x) = x^2$ द्वारा दी गई है : **[JEE(Main)Aug.2021]**

(A) $x^3 - 3xf(x) - 4 = 0$ (B) $x^3 + xf(x) + 12 = 0$

(C) $x^2 + 2xf(x) + 4 = 0$ (D) $x^2 + 2xf(x) - 12 = 0$

Ans. A

Sol. $f(x) + x\,f'(x) = x^2$

Let $y = f(x)$

$$y + x\frac{dy}{dx} = x^2$$

$$\frac{dy}{dx} + \frac{y}{x} = x$$

If $= e^{\int \frac{dx}{x}} = x$

$$y \cdot x = \int x^2 dx = \frac{x^3}{3} + C$$

Passes through (–2, 2)

$$\Rightarrow C = \frac{-4}{3}$$

$$yx = \frac{x^3}{3} - \frac{4}{3}$$

$\Rightarrow x^3 - x\ f(x) - 4 = 0$

18. If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points (0, 1) and (2, β), then β is a root of the equation : **[JEE(Main)Aug.2021]**

यदि अवकल समीकरण $(2x - 10y^3)dy + ydx = 0$ का हल वक्र, बिन्दुओं (0, 1) तथा (2, β) से होकर जाता है, तो β निम्न में से किस समीकरण का एक मूल है : **[JEE(Main)Aug.2021]**

(A) $y^5 - 2y - 2 = 0$ (B) $2y^5 - 2y - 1 = 0$ (C) $y^5 - y^2 - 1 = 0$ (D) $2y^5 - y^2 - 2 = 0$

Ans. C

Sol. $y\dfrac{dx}{dy} = -2x + 10y^3$

$\dfrac{dx}{dy} + x\left(\dfrac{2}{y}\right) = 10y^2$

I.F. $= e^{\int \frac{2}{y}dy} = y^2$

$= e$

Now

$x(IF) = \int Q(IF)dy$

$xy^2 = \int 10y^2(y^2)dy$

$xy^2 = 2y^5 + c$

from (0, 1)

$c = -2$

$xy^2 = 2y^5 - 2$

$xy^5 - xy^2 - 2 = 0$

from (2, B)

$2B^5 - 2B^2 - 2 = 0$

$B^5 - B^2 - 1 = 0$

$y^5 - y^2 - 1 = 0$

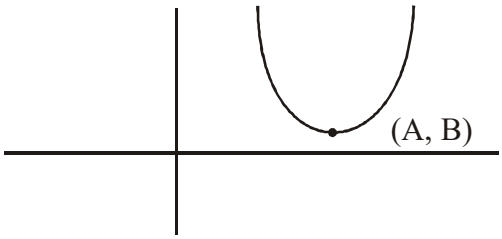
19. A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point (2, –3) from the line $3x + 4y = 5$, is given by : **[JEE(Main)Aug.2021]**

ऐसे परवलयों के कुल, जिनके अक्ष y-अक्ष के समान्तर हैं तथा जिनकी नाभिलम्ब जीवा की लम्बाई बिन्दु (2, –3) की रेखा $3x + 4y = 5$ से दूरी है, को निरूपित करने वाला एक अवकल समीकरण है : **[JEE(Main)Aug.2021]**

(A) $11\dfrac{d^2x}{dy^2} = 10$ (B) $11\dfrac{d^2y}{dx^2} = 10$ (C) $10\dfrac{d^2x}{dy^2} = 11$ (D) $10\dfrac{d^2y}{dx^2} = 11$

Ans. B

Sol.



$$4a = \left|\frac{6-12-5}{5}\right|$$

$$4a = \frac{11}{5}$$

$$(x-A)^2 = \frac{11}{5}(y-B)$$

$$2(x-A) = \frac{11}{5}\frac{dy}{dx}$$

$$2 = \frac{11}{5}\frac{d^2y}{dx^2}$$

$$11\frac{d^2y}{dx^2} = 10$$

20. Let f be a non-negative function in [0, 1] and twice differentiable in (0, 1). If $\int_0^x \sqrt{1-(f'(t))^2}\ dt = \int_0^x f(t)\,dt, 0 \le x \le 1$ and $f(0) = 0$, then $\lim\limits_{x\to 0}\frac{1}{x^2}\int_0^x f(t)\ dt$:

[JEE(Main)Aug.2021]

(A) equals 0 (B) does not exist (C) equals 1 (D) equals $\frac{1}{2}$

माना f, [0, 1] में ऋणेतर तथा (0, 1) में दो बार अवकलनीय है। यदि $\int_0^x \sqrt{1-(f'(t))^2}\ dt = \int_0^x f(t)\,dt, 0 \le x \le 1$ तथा $f(0) = 0$ हैं, तो $\lim\limits_{x\to 0}\frac{1}{x^2}\int_0^x f(t)\ dt$: **[JEE(Main)Aug.2021]**

(A) 0 के बराबर (B) का अस्तित्व नहीं है (C) 1 के बराबर है (D) $\frac{1}{2}$ के बराबर है

Ans. D

Sol. $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)dt$

use leibintz theorem

$$\sqrt{1-(f'(x))^2} = f(x)$$

$$f'(x) = \sqrt{1-(f(x))^2}$$

$$\int \frac{f'(x)}{\sqrt{1-(f(x))^2}}dx = \int dx \Rightarrow \sin^{-1}(f(x)) = x + c$$

$$f(0) = 0 \Rightarrow C = 0$$

$$f(x) = \sin x$$

$$\lim_{x\to 0} \frac{\int_0^x f(t)dt}{x^2}\left(\frac{0}{0}\right)$$

Use L'Hospital Rule

$$\lim_{x\to 0} \frac{f(x)}{2x} = \lim_{x\to 0} \frac{\sin x}{2x} = \frac{1}{2}$$

21. If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, $y(0) = 1$, then $y(1)$ is equal to : **[JEE(Main)Aug.2021]**

यदि $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, $y(0) = 1$ हैं, तो $y(1)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\log_2(2 + e)$ (B) $\log_2(1 + e)$ (C) $\log_2(2e)$ (D) $\log_2(1 + e^2)$

Ans. B

Sol. $\int \frac{2^y dy}{2^y - 1} = \int 2^x dx$

$$\ln(2^y - 1) = 2^x + C$$

$$y(0) = 1 \Rightarrow C = -1$$

$$\ln(2^y - 1) = 2^x - 1$$

$$2^y = 1 + e^{2^x - 1}$$

$$y = \log_2(1 + e^{2^x - 1})$$

$$y(1) = \log_2(1 + e)$$

22. If $y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, $x > 0$, $\phi > 0$, and $y(1) = -1$, then $\phi\left(\frac{y^2}{4}\right)$ is equal to :

[JEE(Main)Aug.2021]

यदि $y\frac{dy}{dx}=x\left[\frac{y^2}{x^2}+\frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, $x>0$, $\phi>0$, तथा $y(1)=-1$ हैं, तो $\phi\left(\frac{y^2}{4}\right)$ बराबर है :

[JEE(Main)Aug.2021]

(A) $2\phi(1)$ (B) $\phi(1)$ (C) $4\phi(2)$ (D) $4\phi(1)$

Ans. D

Sol. $y\frac{dy}{dx}=x\left[\frac{y^2}{x^2}+\frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$

$y=Vx$

$\frac{dy}{dx}=v+x\frac{dv}{dx}$

$vx\left(v+\frac{xdv}{dx}\right)=x\left(v^2+\frac{\phi(v^2)}{\phi'(v^2)}\right)$

$vx\frac{dv}{dx}=\frac{\phi(v^2)}{\phi'(v^2)}$

$\int\frac{v\phi'(v^2)}{\phi(v^2)}dv=\int\frac{dx}{x}$

$\phi(v^2)=u$

$\phi'(v^2)\times 2vdv=du$

$v\phi'(v^2)dv=\frac{du}{2}$

$\frac{1}{2}\int\frac{du}{u}=\int\frac{dx}{x}$

$\frac{1}{2}l\,n\,u=\ln x+c$

$\frac{1}{2}l\,n(\phi(v^2))=\ln x+c$

$$\frac{1}{2}l\,n\left(\phi\left(\frac{y^2}{x^2}\right)\right)=\ln x+c$$

$$\frac{1}{2}l\,n\left(\phi\left(\frac{1}{1}\right)\right)=c$$

$$c=\frac{1}{2}l\,n(\phi(1))$$

$$\frac{1}{2}l\,n\left(\phi\left(\frac{y^2}{x^2}\right)\right)=l\,n\,x+\frac{1}{2}l\,n(\phi(1))$$

$$\frac{1}{2}l\,n\left(\phi\left(\frac{y^2}{4}\right)\right)=l\,n\,2+\frac{1}{2}l\,n(\phi(1))$$

$$l\,n\left(\phi\left(\frac{y^2}{4}\right)\right)=l\,n\,4+l\,n(\phi(1))$$

$$l\,n\left(\phi\left(\frac{y^2}{4}\right)\right)=l\,n(4\phi(1))$$

$$\phi\left(\frac{y^2}{4}\right)=4\phi(1)$$

23. If $\frac{dy}{dx}=\frac{2^x y+2^y\cdot 2^x}{2^x+2^{x+y}\log_e 2}$, $y(0)=0$, then for $y=1$, the value of x lies in the interval :

[JEE(Main)Aug.2021]

यदि $\frac{dy}{dx}=\frac{2^x y+2^y\cdot 2^x}{2^x+2^{x+y}\log_e 2}$, $y(0)=0$ है, तो $y=1$ के लिए x का मान निम्न में से किस अंतराल में है :

[JEE(Main)Aug.2021]

(A) $\left(\frac{1}{2},1\right]$ (B) $(1,2)$ (C) $\left(0,\frac{1}{2}\right]$ (D) $(2,3)$

Ans. B

Sol. $$\frac{dy}{dx}=\frac{2^x y+2^y.2^x}{2^x+2^{x+y}l\,n\,2}$$

$$\frac{dy}{dx}=\frac{2^x(y+2^y)}{2^x(1+2^y l\,n\,2)}$$

$$\int \frac{1+2^y l\,n\,2}{y+2^y}dy = \int dx$$

$$y + 2^y = t$$

$$(1+2^y l\,n\,2)dy = dt$$

$$\int \frac{dt}{t} = \int dx$$

$$l nt = x + c$$

$$l n(y + 2^y) = x + c$$

$$l n(0 + 1) = 0 + c$$

$$c = 0$$

$$l n(y + 2^y) = x$$

$$l n(1 + 2) = x$$

$$x = l n3$$

$$x \in (1, 2)$$

24. If $y = y(x)$ is the solution curve of the differential equation $x^2dy + \left(y - \frac{1}{x}\right)dx = 0$; $x > 0$, and $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to : **[JEE(Main)Aug.2021]**

यदि $y = y(x)$ अवकल समीकरण $x^2dy + \left(y - \frac{1}{x}\right)dx = 0$; $x > 0$ का हल वक्र है तथा $y(1) = 1$ है, तब $y\left(\frac{1}{2}\right)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $3 + \frac{1}{\sqrt{e}}$ (C) $3 - e$ (D) $3 + e$

Ans. C

Sol. $x^2dy = \left(\frac{1}{x} - y\right)dx$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x^3} - \frac{1}{x^2}y$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x^2}y = \frac{1}{x^3}$$

I.F. $= e^{\int \frac{1}{x^2}dx} = e^{-\frac{1}{x}}$

Hence,

$$y.e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}}.\frac{1}{x^3}.dx + c$$

$$y e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}}.\frac{1}{x}\frac{1}{x^2}dx + c$$

Put $t = \frac{-1}{x}$

$$y e^{-\frac{1}{x}} = \int e^{t}(-t)dt + c$$

$$y e^{-\frac{1}{x}} = -\int e^{t}dt + c$$

$$y e^{-\frac{1}{x}} = -[te^{t} - e^{t}] + c$$

$$y e^{-\frac{1}{x}} = \frac{1}{x}e^{-\frac{1}{x}} + e^{-\frac{1}{x}} + c$$

$$y = \frac{1}{x} + 1 + c e^{\frac{1}{x}}$$

$\because y(1) = 1 \qquad 1 = 1 + 1 + c e$

$$c = -\frac{1}{e}$$

$$y = \frac{1}{x} + 1 - \frac{1}{e}\cdot e^{\frac{1}{x}}$$

$$y\left(\frac{1}{2}\right) = 2 + 1 - \frac{1}{e}.e^{2} \quad = \quad 3 - e$$

25. If $x\,\phi(x) = \int_5^x \left(3t^2 - 2\phi'(t)\right)dt$, $x > -2$, and $\phi(0) = 4$, then $\phi(2)$ is _____. **[JEE(Main)Aug.2021]**

यदि $x\,\phi(x) = \int_5^x \left(3t^2 - 2\phi'(t)\right)dt$, $x > -2$, तथा $\phi(0) = 4$ है, तो $\phi(2)$ बराबर है _____

[JEE(Main)Aug.2021]

Ans. 4

Sol. $3x.\phi(x) = \int_5^x (3t^2 - 2\phi'(t))dt$

$\Rightarrow x.\phi'(x) + \phi(x) = 3x^2 - 2\phi'(x)$

$$\Rightarrow \phi'(x)+\frac{1}{x+2}\phi(x)=\frac{3x^2}{x+2}$$

Let $\phi(x)=y$

$$\Rightarrow \frac{dy}{dx}+\frac{y}{x+2}=\frac{3x^2}{x+2}$$

I.F. $=e^{\int\frac{1}{x+2}dx}=x+2$

$$y.(x+2)=\int\frac{3x^2}{(x+2)}.(x+2)dx$$

$$y(x+2)=x^3+C$$

$$y(0)=4\Rightarrow C=8$$

$$\phi(x)=\frac{x^3+8}{x+2}$$

$$\phi(2)=\frac{16}{4}=4$$

26. If $x=x(y)$ is the solution of the differential equation $y\frac{dx}{dy}=2x+y^3(y+1)e^y, x(1)=0$; then $x(e)$ is equal to :

[JEE(Main)June2022]

यदि अवकल समीकरण $y\frac{dx}{dy}=2x+y^3(y+1)e^y, x(1)=0$ का हल $x=x(y)$ है, तो $x(e)$ बराबर है :

[JEE(Main)June2022]

(A) $e^3(e^e-1)$ (B) $e^e(e^3-1)$ (C) $e^2(e^e+1)$ (D) $e^e(e^2-1)$

Ans. A

Sol. Linear differential equation

$$y\frac{dx}{dy}-2x=y^3(y+1)e^y$$

$$\frac{dx}{dy}-\frac{2}{y}x=y^2(y+1)e^y$$

I.f$=e^{\int-\frac{2}{y}dy}=\frac{1}{y^2}$

$$\int d\left(\frac{x}{y^2}\right)=\int(y+1)e^y dy$$

$$\frac{x}{y^2} = ye^y + c$$

$x(1) = 0 \Rightarrow c = -e$

$$\frac{x}{y^2} = ye^y - e$$

put $y = e$

$$\frac{x}{e^2} = ee^e - e = e\left(e^e - 1\right)$$

$$x = e^3\left(e^e - 1\right)$$

27. A particle is moving in the xy-plane along a curve C passing through the point (3, 3). The tangent to the curve C at the point P meets the x-axis at Q. If the y-axis bisects the segment PQ, then C is a parabola with :

[JEE(Main)June2022]

(A) length fo latus rectum 3 (B) length of latus rectum 6

(C) focus $\left(\frac{4}{3}, 0\right)$ (D) focus $\left(0, \frac{3}{4}\right)$

xy-समतल में कण, बिन्दु (3, 3) से होकर जाने वाले एक वक्र C के अनुदिश चल रहा है। माना वक्र C के बिन्दु P पर स्पर्श रेखा, x-अक्ष को बिन्दु Q पर मिलती है। यदि रेखाखण्ड PQ को y-अक्ष समद्विभाजित करता है, तो C एक परवलय है जिसकी :

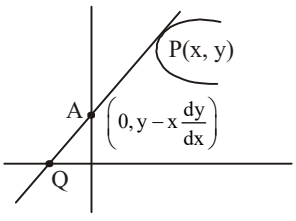
[JEE(Main)June2022]

(A) नाभिलंब जीवा की लम्बाई 3 है (B) नाभिलंब जीवा की लम्बाई 6 है

(C) नाभी $\left(\frac{4}{3}, 0\right)$ है (D) नाभी $\left(0, \frac{3}{4}\right)$ है

Ans. A

Sol.



tangent at 'P'

$$Y - y = \frac{dy}{dx}(X - x)$$

$$A\left(-y\frac{dx}{dy} + x, 0\right), Q\left(x - \frac{y}{dy/dx}, 0\right), P(x, y)$$

Since A is mid point of segment PQ

$$\frac{x - y\frac{dx}{dy} + x}{1+1} = 0$$

$$y\frac{dx}{dy} = 2x$$

$$\int 2\frac{dy}{y} = \int \frac{dx}{x}$$

$2\,l\text{n}y = l\text{n}x + l\text{n}c$

$2l\text{n}y = l\text{n}cx$

$y^2 = cx$

Curve passing through (3, 3) then

$9 = 3c$

$c = 3$

$y^2 = 3x$

Length of latus rectum is 3.

28. The slope of normal at any point (x, y), $x > 0$, $y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$. If the curve passes through the point (1, 1), then e.y(e) is equal to : **[JEE(Main)June2022]**

वक्र $y = y(x)$ के किसी भी बिन्दु (x, y), $x > 0$, $y > 0$ पर अभिलंब की प्रवणता $\frac{x^2}{xy - x^2y^2 - 1}$ द्वारा दी गई है। यदि यह वक्र बिन्दु (1, 1) से होकर जाता है, तो e.y(e) बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1 - \tan(1)}{1 + \tan(1)}$ (B) $\tan(1)$ (C) 1 (D) $\frac{1 + \tan(1)}{1 - \tan(1)}$

Ans. D

Sol. $$-\frac{dx}{dy} = \frac{x^2}{xy - x^2y^2 - 1}$$

$$\frac{dy}{dx} = -\left(\frac{xy - x^2y^2 - 1}{x^2}\right)$$

$$\frac{dy}{dx} = \left(\frac{x^2y^2 - xy + 1}{x^2}\right)$$

$xy = v$

$$y + x\frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{1}{x}\left(\frac{dv}{dx} - y\right)$$

$$\left(\frac{dv}{dx}-y\right)=\frac{v^2-v+1}{x}$$

$$\frac{dv}{dx}-\frac{v}{x}=\frac{v^2-v+1}{x}$$

$$x\frac{dv}{dx}=v^2+1$$

$$\int\frac{dv}{v^2+1}=\int\frac{dx}{x}$$

$\tan^{-1}(v) = \ell nx + c$

$\tan^{-1}(xy) = \ell nx + c$

passing through (1, 1)

$\tan^{-1}1 = 0 + c$

$\tan^{-1}(xy) = \ell nx + \tan^{-1}(1)$

$xy = \tan(\ell nx + \tan^{-1}(1))$

$$y=\frac{1}{x}\tan\left(\ell nx+\frac{\pi}{4}\right)$$

$$y(e)=\frac{1}{e}\tan\left(1+\frac{\pi}{4}\right)$$

now

$$e.\,y(e)=\frac{1+\tan 1}{1-\tan 1}$$

29. Let $y = y(x)$ be the solution of the differential equation $(x+1)y'-y=e^{3x}(x+1)^2$, with $y(0)=\frac{1}{3}$. Then, the point $x=-\frac{4}{3}$ for the curve $y = y(x)$ is : **[JEE(Main)June2022]**

(A) not a critical point (B) a point of local minima

(C) a point of local maxima (D) a point of inflection

माना अवकल समीकरण $(x+1)y'-y=e^{3x}(x+1)^2$, $y(0)=\frac{1}{3}$, का हल $y = y(x)$ है। तो वक्र $y = y(x)$ के लिए बिन्दु $x=-\frac{4}{3}$ एक : **[JEE(Main)June2022]**

(A) क्रांतिक बिन्दु नहीं है (B) स्थानीय निम्ननिष्ठ बिन्दु है

(C) स्थानीय उच्चिष्ठ बिन्दु है (D) नति परिवर्तन बिन्दु है

Ans. B

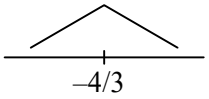
Sol. $(x+1)dy-ydx=e^{3x}(x+1)^2$

$$\frac{(x+1)dy-ydx}{(x+1)^2}=e^{3x}$$

$$d\left(\frac{y}{x+1}\right)=e^{3x}\Rightarrow\frac{y}{x+1}=\frac{e^{3x}}{3}+C$$

$$\left(0,\frac{1}{3}\right)\Rightarrow C=0\Rightarrow y=\frac{(x+1)e^{3x}}{3}$$

$$\frac{dy}{dx}=\frac{1}{3}\left((x+1)3e^{3x}+e^{3x}\right)=\frac{e^{3x}}{3}(3x+4)$$



Clearly, $x=\frac{-4}{3}$ is point of local minima

30. If the solution curve $y=y(x)$ of the differential equation $y^2dx+(x^2-xy+y^2)dy=0$, which passes through the point $(1, 1)$ and intersects the line $y=\sqrt{3}x$ at the point $\left(\alpha,\sqrt{3}\alpha\right)$, then value of $\log_e\left(\sqrt{3}\alpha\right)$ is equal to :

[JEE(Main)June2022]

यदि अवकल समीकरण $y^2dx+(x^2-xy+y^2)dy=0$ का हल वक्र $y=y(x)$ है, जो बिन्दु $(1, 1)$ से होकर जाता है तथा रेखा $y=\sqrt{3}x$ को बिन्दु $\left(\alpha,\sqrt{3}\alpha\right)$ पर काटता है, तो $\log_e\left(\sqrt{3}\alpha\right)$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

Ans. C

Sol. $y^2dx-xydy=-\left(x^2+y^2\right)dy$

$$y(ydx-xdy)=-\left(x^2+y^2\right)dy$$

$$\frac{xdy-ydx}{x^2}=\left(1+\frac{y^2}{x^2}\right)\frac{dy}{y}$$

$$\Rightarrow\frac{d(y/x)}{1+\frac{y^2}{x^2}}=\frac{dy}{y}$$

$$\Rightarrow\tan^{-1}\left(\frac{y}{x}\right)=\ln y+C\Rightarrow\text{passes through}\,(1,1)\Rightarrow c=\frac{\pi}{4}$$

$$(\alpha,\sqrt{3}\alpha)\Rightarrow\frac{\pi}{3}=\ln(\sqrt{3}\alpha)+\frac{\pi}{4}$$

$\therefore \quad \ln(\sqrt{3}\alpha) = \frac{\pi}{12}$

31. If $y = y(x)$ is the solution of the differential equation $2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then $y(1)$ is equal to : **[JEE(Main)June2022]**

यदि $y = y(x)$, अवकल समीकरण $2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$, $y(e) = \frac{e}{3}$ का हल है, तो $y(1)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Ans. B

Sol. $2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$

$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y}.\frac{1}{x} = \frac{-3}{2x^2}$

Put $-\frac{1}{y} = z$

$\frac{1}{y^2}.\frac{dy}{dx} = \frac{dz}{dx}$

$\frac{dz}{dx} + \frac{1}{x}.z = \frac{-3}{2x^2}$

I.F. $= e^{\int\frac{1}{x}dx}$

Solution $z.x = \int\frac{-3}{2x^2}.x\,dx$

$-\frac{1}{y}.x = \frac{-3}{2}\ln x - C$

$\frac{x}{y} = \frac{3}{2}\ln x + C$

$y(e) = \frac{e}{3} \Rightarrow C = \frac{3}{2}$

$y = \frac{2}{3}.\frac{x}{(\ln x + 1)}$

$y(1) = \frac{2}{3}$

32. If $y=(x)$ is the solution of the differential equation $x\frac{dy}{dx}+2y=xe^x, y(1)=0$ then the local maximum value of function $z(x)=x^2y(x)-e^x, x\in R$ is: **[JEE(Main)June2022]**

यदि अवकल समीकरण $x\frac{dy}{dx}+2y=xe^x, y(1)=0$ का हल $y=y(x)$ है, तो फलन $z(x)=x^2y(x)-e^x, x\in R$ का स्थानीय अधिकतम मान है : **[JEE(Main)June2022]**

(A) $1-e$ (B) 0 (C) 1/2 (D) $\frac{4}{e}-e$

Ans. D

Sol. $x\frac{dy}{dx}+zy=xe^x$

$\frac{dy}{dx}+\left(\frac{2}{x}\right)y=e^x$

I. $F=e^{\int\frac{2dx}{x}}=e^{2\ell nx}e^{\ell nx^2}=x^2$

$yx^2=\int e^4\times x^2dx$

$yx^2=x^2\times e^x-\int 2x\times e^xdy$

$yx^2=x^2e^x-2\left[xe^x-\int e^xdx\right]$

$x^2y=x^2e^x-2xe^x+2e^x+c$

$1\times 0=1\times e-\frac{2}{e}+\frac{2}{e}+c$

$c=-e$

$x^2y(x)=x^2e^x-2xe^x+2e^x-e$

$z(x)=x^2e^x-2xe^x+2e^x-e-e^x$

$z(x)=x^2e^x-2xe^x+e^x-e$

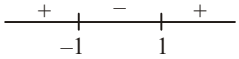
$z'(x)=x^2e^x+2xe^x-2\left(xe^x+e^x\right)+e^x$

$z'(x)=x^2e^x+2xe^x-2xe^x-2e^x+e^x$

$z'(x)=x^2e^x-e^x=0$

$\Rightarrow e^x(x^2-1)=0$

$\Rightarrow x=\pm1$



$z(-1)=1\times e^{-1}-2(-1)\times e^{-1}+e^{-1}-e$

$=\frac{1}{e}+\frac{2}{e}+\frac{1}{e}-e$

$= \frac{4}{e} - e$

33. If the solution of the differential equation $\frac{dy}{dx} + e^x\left(x^2-2\right)y = \left(x^2-2x\right)\left(x^2-2\right)e^{2x}$ satisfies y(0) = 0, then the value of y(2) is : **[JEE(Main)June2022]**

यदि अवकल समीकरण $\frac{dy}{dx} + e^x\left(x^2-2\right)y = \left(x^2-2x\right)\left(x^2-2\right)e^{2x}$ का हल y(0) = 0 को संतुष्ट करता है, तो y(2) का मान है : **[JEE(Main)June2022]**

(A) – 1 (B) 1 (C) 0 (D) e

Ans. C

Sol. $\frac{dy}{dx} + e^x\left(x^2-2\right)y = \left(x^2-2x\right)\left(x^2-2\right)e^{2x}$

IF = $e^{\int e^x\left(x^2-2\right)dx}$

Solution:

$y.e^{e^x\left(x^2-2x\right)} = \int e^{e^x\left(x^2-2x\right)}\left(x^2-2x\right)\left(x^2-2\right)e^{2x}dx$

$t = e^x\left(x^2-2x\right)$

$dt = e^x\left(x^2-2\right)dx$

$ye^t = \int e^t t.dt$

$ye^t = te^t - e^t + c$

$ye^{e^x\left(x^2-2x\right)} = e^{e^x\left(x^2-2x\right)}\left(e^x\left(x^2-2x\right)-1\right)+c$

y (0) = 0

$0 = e^0(e^0 0 - 1) + c$

c = 1

$ye^{e^x\left(x^2-2x\right)} = e^{e^x\left(x^2-2x\right)}\left(e^x\left(x^2-2x\right)-1\right)+1$

y(2) : ye

$ye^{e^2(0)} = e^{e^2 0}\left(e^2 0 - 1\right)+1$

y = 0

34. Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, where a,b,c are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11, 6) from the circle is : **[JEE(Main)June2022]**

माना $\frac{dy}{dx}=\frac{ax-by+a}{bx+cy+a}$, a, b, c अचर हैं, बिन्दु (2, 5) से होकर जाने वाले वृत्त को निरुपित करता है। तो बिन्दु (11, 6) की इस वृत्त से न्यूनतम दूरी है : **[JEE(Main)June2022]**

(A) 10 (B) 8 (C) 7 (D) 5

Ans. B

Sol. $\frac{dy}{dx}=\frac{ax-by+a}{bx+cy+a}$

$bxdy+cydy+ady=axdx-bydx+adx$

$b\int(xdy+ydx)+\int cydy+\int ady=\int axdx+\int adx$

$b\,xy+\frac{cy^2}{2}+ay=\frac{ax^2}{2}+ax+k$

for a circle $b=0$

$c=-a$

$\frac{a}{2}x^2+\frac{a}{2}y^2+ax-ay+k=0$

$x^2+y^2+2x-2y+k=0$

(2, 5) $k=-23$

$x^2+y^2+2x-2y-23=0$

centre (–1, 1) $r=\sqrt{1+1+23}=5$

Shortest distance from p(11, 6) = |op – r|

$=\left|\sqrt{(12)^2+(5)^2}-5\right|$

$=|13-5|$

$=8$

35. If $\frac{dy}{dx}+\frac{2^{x-y}\left(2^y-1\right)}{2^x-1}=0$, x, y > 0, y(1) = 1, then y(2) is equal to : **[JEE(Main)June2022]**

यदि $\frac{dy}{dx}+\frac{2^{x-y}\left(2^y-1\right)}{2^x-1}=0$, x, y > 0, y(1) = 1 हैं, तो y(2) बराबर है : **[JEE(Main)June2022]**

(A) $2+\log_2 3$ (B) $2+\log_2 2$ (C) $2-\log_2 2$ (D) $2-\log_2 3$

Ans. D

Sol. $\frac{dy}{dx}=\frac{-2^{x-y}\left(2^y-1\right)}{2^x-1}$

$\frac{dy}{dx}=\frac{-2^x.2^{-y}\left(2^y-1\right)}{2^x-1}$

$$\frac{dy}{dx}=\frac{-2^x\left(1-2^{-y}\right)}{2^x-1}$$

$$\int\frac{dy}{1-2^{-y}}=-\int\frac{2^x}{2^x-1}dx$$

$$\int\frac{2^y dx}{2^y-1}=-\int\frac{2^x}{2^x-1}dx$$

Let $2^y-1=t$ $\qquad$ $2^x-1=u$

$2^y\,\ell n2\,dy=dt$ $\qquad$ $2^x\,ln2\,dx=du$

$$\int\frac{1}{t}dt=-\int\frac{1}{u}du$$

$ln\,t=-ln\,u+ln\,c$

$$t=\frac{c}{u}$$

$$2^y-1=\frac{c}{2^x-1}$$

Now $x=1,$ $\qquad$ $y=1$

$\Rightarrow c=1$

$$2^y-1=\frac{1}{2^x-1}$$

Now put $x=2$

$$2^y-1=\frac{1}{4-1}$$

$$2^y=1+\frac{1}{3}=\frac{4}{3}$$

$$y=\log_2\frac{4}{3}=2-\log_2 3$$

36. If the solution curve of the differential equation $((\tan^{-1}y)-x)dy=(1+y^2)dx$ passes through the point $(1,0)$, then the abscissa of the point on the curve whose ordinate is $\tan(1)$, is : **[JEE(Main)June2022]**

यदि अवकल समीकरण $((\tan^{-1}y)-x)dy=(1+y^2)dx$ का हल वक्र बिन्दु $(1,0)$ से होकर जाता है, तो वक्र पर उस बिन्दु, जिसकी कोटि $\tan(1)$ है, का भुज है : **[JEE(Main)June2022]**

(A) 2e $\qquad$ (B) 2/e $\qquad$ (C) 2 $\qquad$ (D) 1/e

Ans. B

Sol. $$\tan^{-1}y-x=\left(1+y^2\right)\frac{dx}{dy}$$

$$\Rightarrow\frac{dx}{dy}=\frac{\tan^{-1}y}{1+y^2}-\frac{x}{1+y^2}$$

$$\Rightarrow \frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}$$

LDE in 'x'

$$\text{I.F.}=e^{\int\frac{dy}{1+y^2}}=e^{\tan^{-1}y}$$

$$x\left(e^{\tan^{-1}y}\right)=\int\frac{\tan^{-1}y}{1+y^2}.e^{\tan^{-1}y}dy+c$$

$$xe^{\tan^{-1}y}=\left(\tan^{-1}y\right)e^{\tan^{-1}y}-e^{\tan^{-1}y}+c$$

put x = 1, y = 0 put y = tan 1

$1 = 0 - 1 + c$ $xe^1 = e - e + 2$

$\Rightarrow c = 2$ $x = \frac{2}{e}$

37. Let the solution curve y = y(x) of the differential equation $\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]x\frac{dy}{dx}=x+\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]y$ pass through the points (1, 0) and $(2\alpha, \alpha)$, $\alpha > 0$. Then α is equal to : **[JEE(Main)June2022]**

माना अवकल समीकरण $\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]x\frac{dy}{dx}=x+\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]y$ का हल वक्र y = y(x) बिन्दुओं (1, 0) तथा $(2\alpha, \alpha)$, $\alpha > 0$ से होकर जाता है, तो α का मान बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$ (B) $\frac{1}{2}\exp\left(\frac{\pi}{3}+e-1\right)$

(C) $\exp\left(\frac{\pi}{6}+\sqrt{e}+1\right)$ (D) $2\exp\left(\frac{\pi}{3}+\sqrt{e}-1\right)$

Ans. A

Sol. $$\left(\frac{x}{\sqrt{x^2-y^2}}+e^{(y/x)}\right)\left(x\frac{dy}{dx}-y\right)=\alpha$$

$$\Rightarrow\left(\frac{1}{\sqrt{1-\left(\frac{y}{x}\right)^2}}+e^{(y/x)}\right)d\left(\frac{y}{x}\right)=\frac{1}{x}dx$$

Inteqrating both sides

$$\sin^{-1}\left(\frac{y}{x}\right)+e^{(y/x)}=\ln x+c$$

as y (1) = 0 so c = 1

$$\sin^{-1}\left(\frac{y}{x}\right)+e^{(y/x)}=\ln x+1$$

$(2\alpha, \alpha)$ will satisfy it

$$\frac{\pi}{6}+\sqrt{e}-1=\ln(2\alpha)$$

$$\Rightarrow \alpha=\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$$

38. Let $y=y(x)$ be the solution of the differential equation $x\left(1-x^2\right)\frac{dy}{dx}+\left(3x^2y-y-4x^3\right)=0,\ x>1$, with $y(2)=-2$. Then $y(3)$ is equal to : **[JEE(Main)June2022]**

माना अवकल समीकरण $x\left(1-x^2\right)\frac{dy}{dx}+\left(3x^2y-y-4x^3\right)=0,\ x>1$ का हल $y=y(x)$ है तथा $y(2)=-2$ तो $y(3)$ बराबर है : **[JEE(Main)June2022]**

(A) –18 (B) –12 (C) –6 (D) –3

Ans. A

Sol. $$\frac{dy}{dx}-\frac{y\left(3x^2-1\right)}{x\left(x^2-1\right)}=\frac{4x^3}{x\left(1-x^2\right)}$$

$$\text{If}= e^{-\int\left(\ln\left(x^2-1\right)+\ln x\right)}=\frac{1}{x\left(x^2-1\right)}$$

$$\frac{y}{x\left(x^2-1\right)}=\int\frac{4x^3}{x\left(1-x^2\right)}\cdot\frac{1}{x\left(x^2-1\right)}dx=\int\frac{-4x}{\left(x^2-1\right)^2}dx=\frac{2}{x^2-1}+c$$

$\Rightarrow y = 2x + cx(x^2 - 1)$

Now $y(2) = -2 \Rightarrow c = -1$

So $y = 2x - x(x^2 - 1)$

$y(3) = 2 \times 3 - 3(8) = 6 - 24 = -18$

39. Let $x = x(y)$ be the solution of the differential equation $2y\,e^{\frac{x}{y^2}}dx+\left(y^2-4xe^{\frac{x}{y^2}}\right)dy=0$ such that $x(1) = 0$. Then, $x(e)$ is equal to :

माना अवकल समीकरण $2y\, e^{\frac{x}{y^2}}dx+\left(y^2-4xe^{\frac{x}{y^2}}\right)dy=0$ का हल x = x (y) इस प्रकार है कि x(1) = 0 है तो x(e) बराबर है :

(A) $e\log_e(2)$ (B) $-e\log_e(2)$ (C) $e^2\log_e(2)$ (D) $-e^2\log_e(2)$

Ans. D

Sol. $2.ye^{\left(\frac{x}{y^2}\right)}.dx+\left(y^2-4x.e^{\left(\frac{x}{y^2}\right)}\right).dy=0$

$$\Rightarrow 2e^{\left(\frac{x}{y^2}\right)}\left[y.dx-2xdy\right]=-y^2.dy$$

$$\because d\left(\frac{x}{y^2}\right)=\frac{y^2.dx-x.2y.dy}{y^4}$$

So multiple both sides by "$\frac{y}{y^4}$"

$$\Rightarrow 2e^{\left(\frac{x}{y^2}\right)}\left[\frac{y^2.dx-2xy.dy}{y^4}\right]=-\frac{dy}{y}$$

$$\Rightarrow \int 2e^{\left(\frac{x}{y^2}\right)}.d\left(\frac{x}{y^2}\right)=\int-\frac{dy}{y}$$

$$\Rightarrow 2e^{\left(\frac{x}{y^2}\right)}=-\ell n\, y+c \qquad \text{.....(1)}$$

given $x(1)=0\Rightarrow x=0, y=1\Rightarrow c=2.$

$$\Rightarrow 2e^{\left(\frac{x}{y^2}\right)}=-\ell n\, y+2$$

for x(e)

$$\Rightarrow 2e^{\left(\frac{x}{e^2}\right)}=-\ell n\, e+2=-1+2$$

$$\Rightarrow 2e^{\frac{x}{e^2}}=1 \qquad \Rightarrow e^{\left(\frac{x}{e^2}\right)}=\frac{1}{2}$$

$$\frac{x}{e^2}=\ell n\left(\frac{1}{2}\right) \qquad \Rightarrow x=e^2\times-\ell n\, 2$$

40. Let the slope of the tangent to a curve y = f(x) at (x, y) be given by 2 tan x(cos x – y). If the curve passes through the point $\left(\frac{\pi}{4},0\right)$, then the value of $\int_0^{\frac{\pi}{2}} y\,dx$ is equal to : **[JEE(Main)June2022]**

माना एक वक्र $y = f(x)$ के बिन्दु (x, y) पर स्पर्श रेखा की प्रवणता $2\tan x(\cos x - y)$ है। यदि यह वक्र बिन्दु $\left(\frac{\pi}{4}, 0\right)$ से होकर जाता है, तो $\int_0^{\frac{\pi}{2}} y\,dx$ का मान बराबर : **[JEE(Main)June2022]**

(A) $\left(2-\sqrt{2}\right)+\frac{\pi}{\sqrt{2}}$ (B) $2-\frac{\pi}{\sqrt{2}}$ (C) $\left(2+\sqrt{2}\right)+\frac{\pi}{\sqrt{2}}$ (D) $2+\frac{\pi}{\sqrt{2}}$

Ans. B

Sol. $\frac{dy}{dx} = 2\tan x\left(\cos x - y\right)$

$\frac{dy}{dx} + 2\tan x.y = 2.\sin x.$ [its a linear D.E.]

I.F. $= e^{\int 2\tan x.dx} = \sec^2 x$

$\therefore$ solution fo differential equation will be

$y.(\text{I.f}) = \int Q.(\text{IF}).dx$

$y.\sec^2 x = \int 2.\sin x.\sec^2 x.dx$

$y.\sec^2 x = 2\sec x + c$ ——————(1)

Curve is passing through $\left(\frac{\pi}{4}, 0\right)$

$0 = 2\sqrt{2} + c \qquad \Rightarrow c = -2\sqrt{2}$

from equation (1)

$y = 2.\cos x - 2\sqrt{2}\cos^2 x$

$\int_0^{\frac{\pi}{2}} y.dx = \int_0^{\frac{\pi}{2}} \left(2\cos x - 2\sqrt{2}\cos^2 x\right).dx$

$= 2 - 2\sqrt{2}\times\frac{\pi}{4} = 2 - \frac{\pi}{\sqrt{2}}$

41. Let the solution curve of the differential equation $x\frac{dy}{dx} - y = \sqrt{y^2+16x^2}$, $y(1) = 3$ be $y = y(x)$. Then $y(2)$ is equal to : **[JEE(Main)June2022]**

माना अवकल समीकरण $x\frac{dy}{dx} - y = \sqrt{y^2+16x^2}$, $y(1) = 3$ का हल वक्र $y = y(x)$ है। तो $y(2)$ का मान है : **[JEE(Main)June2022]**

(A) 15 (B) 11 (C) 13 (D) 17

Ans. A

Sol. $x\frac{dy}{dx}-y=\sqrt{y^2+16x^2}$

$$\frac{dy}{dx}=\frac{y}{x}+\sqrt{\frac{y^2}{x^2}+16} \qquad \text{.....(1)}$$

Put in (1) $\frac{y}{x}=\mu \Rightarrow \frac{dy}{dx}=\mu+\frac{xd\mu}{dx}$

$$\mu+\frac{xd\mu}{dx}=\mu+\sqrt{\mu^2+16}$$

$$\frac{d\mu}{\sqrt{\mu^2+16}}=\frac{dx}{x}$$

integrate

$$\ln\left(\mu+\sqrt{\mu^2+16}\right)=\ln x+\ln c$$

$$\frac{y}{x}+\frac{\sqrt{y^2+16x^2}}{x}=cx$$

$$y+\sqrt{y^2+16x^2}=cx^2 \qquad \text{.....(2)}$$

given that x = 1 y = 3

$3 + 5 = c \Rightarrow c = 8$

from (2) $y+\sqrt{y^2+16x^2}=8x^2$

put x = 2

then y = 15

42. If y = y(x) is the solution of the differential equation $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$ and y(0) = 0, then $6\left(y'(0)+\left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ is equal to : **[JEE(Main)June2022]**

यदि अवकल समीकरण $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$, $y(0)=0$ का हल $y = y(x)$ है, तो $6\left(y'(0)+\left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ बराबर है : **[JEE(Main)June2022]**

(A) 2 (B) –2 (C) –4 (D) –1

Ans. C

Sol. Given $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$

$$\int \frac{dy}{1+y^2} = -\int \frac{2e^x}{1+e^{2x}}dx = -\int \frac{2e^x}{1+\left(e^x\right)^2}dx$$

let $e^x = t \qquad \Rightarrow e^x\,dx = dt$

$$\Rightarrow \int \frac{dy}{1+y^2} = -2\int \frac{1}{1+t^2}dt$$

$$\Rightarrow \tan^{-1} y = -2\tan^{-1} t + c$$

$$\Rightarrow \tan^{-1} y = -2\tan^{-1} e^x + c$$

Given at $x = 0, y = 0$

$$\Rightarrow c = \frac{\pi}{2}$$

So, $\tan^{-1} y = -2\tan^{-1} e^x + \frac{\pi}{2}$

$$\tan(\tan^{-1} y) = \tan\left(\frac{\pi}{2} - 2\tan^{-1} e^x\right)$$

$$\boxed{y = \cot\left(2\tan^{-1} e^x\right)}$$

$$\frac{dy}{dx} = -\cos ec^2\left(2\tan^{-1} e^x\right)\left(\frac{2e^x}{1+e^{2n}}\right)$$

$$y'(0) = \left.\frac{dy}{dx}\right|_{x=0} = \frac{-2}{2} = -1$$

$$y\left(\ell n\sqrt{3}\right) = \cot\left(2\tan^{-1} e^{\log_e \sqrt{3}}\right)$$

$$y\left(\ell n\sqrt{3}\right) = \cot\left(2\tan^{-1}\sqrt{3}\right)$$

$$y\left(\ell n\sqrt{3}\right) = \cot 2\times\frac{\pi}{3} = \cot\frac{2\pi}{3} = \frac{-1}{\sqrt{3}}$$

$$\therefore 6\left(y'(0) + \left(y\left(\ell n\sqrt{3}\right)\right)^2\right) = 6\left[-1 + \left(\frac{-1}{\sqrt{3}}\right)^2\right]$$

$$= 6\left(-1 + \frac{1}{3}\right) = -4$$

43. Let the solution curve $y = y(x)$ of the differential equation $(4 + x^2)dy - 2x(x^2 + 3y + 4)dx = 0$ pass through the origin. Then $y(2)$ is equal to _____. **[JEE(Main)June2022]**

माना अवकल समीकरण $(4 + x^2)dy - 2x(x^2 + 3y + 4)dx = 0$ का हल वक्र $y = y(x)$ मूल बिन्दु से होकर जाता है। तो $y(2)$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 12

Sol. $(4 + x^2)dy - 2x(x^2 + 3y + 4)dx$

$$\left(x^2+4\right)\frac{dy}{dx} = 2x^3 + 6xy + 8x$$

$$\left(x^2+4\right)\frac{dy}{dx} - 6xy = 2x^3 + 8x$$

$$\frac{dy}{dx} - \frac{6x}{x^2+4}y = \frac{2x^3+8x}{x^2+y}$$

L.I. $\frac{dy}{dx} + py = \phi$

$$p = \frac{-6x}{x^2+4} \qquad \phi = \frac{2x^3+8x}{x^2+4}$$

$$\text{I.F.} = e^{-\int \frac{6x}{x^2+4}dx} = e^{-3\log_e\left(x^2+4\right)}$$

$$= e^{\log_e\left(x^2+4\right)^{-3}} = \frac{1}{\left(x^2+4\right)^3}$$

Sol.

$$y.\frac{1}{\left(x^2+4\right)^3} = \int \frac{2x^3+8x}{\left(x^2+4\right)^3\left(x^2+4\right)}dx$$

$$\frac{y}{\left(x^2+4\right)^3} = \int \frac{2x\left(x^2+4\right)}{\left(x^2+4\right)^3\left(x^2+4\right)}dx$$

$x^2 + 4 = t$

$2xdx = dt$

$$\frac{y}{\left(x^2+4\right)^3} = \int \frac{dt}{t^3}$$

$$\frac{y}{\left(x^2+4\right)^3} = \frac{-1}{2\left(x^2+4\right)^2} + C$$

passes through origin (0, 0)

$$0 = \frac{-1}{2\times 16} + C$$

$$\frac{y}{\left(x^2+4\right)^3}=\frac{-1}{2\left(x^2+4\right)^2}+\frac{1}{32}$$

$$y=\frac{-\left(x^2+4\right)}{2}+\frac{\left(x^2+4\right)^3}{32}$$

$$y(2)=-\frac{8}{2}+\frac{8\times8\times8}{32}=12$$

44. Let $S=(0,2\pi)-\left\{\frac{\pi}{2},\frac{3\pi}{4},\frac{3\pi}{2},\frac{7\pi}{4}\right\}$. Let y = y(x), $x \in S$ be the solution curve of the differential equation $\frac{dy}{dx}=\frac{1}{1+\sin 2x}, y\left(\frac{\pi}{4}\right)=\frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve $y=y(x)$ with the curve $y=\sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to _____. **[JEE(Main)June2022]**

माना $S=(0,2\pi)-\left\{\frac{\pi}{2},\frac{3\pi}{4},\frac{3\pi}{2},\frac{7\pi}{4}\right\}$ है। माना अवकल समीकरण $\frac{dy}{dx}=\frac{1}{1+\sin 2x}, y\left(\frac{\pi}{4}\right)=\frac{1}{2}$ का हल वक्र $y = y(x), x \in S$ है। यदि वक्र $y = y(x)$ के वक्र $y=\sqrt{2}\sin x$ से सभी प्रतिच्छेदन बिंदुओं के भुजों का योग $\frac{k\pi}{12}$ है, तो k बराबर है _____ **[JEE(Main)June2022]**

Ans. 42

Sol. $$\frac{dy}{dx}=\frac{1}{1+\sin 2x}$$

$$\int dy=\int\frac{dx}{(\sin x+\cos x)^2}$$

$$\int dy=\int\frac{\sec^2 x\,dx}{(1+\tan x)^2}$$

$$y(x)=-\frac{1}{1+\tan x}+C$$

$$y\left(\frac{\pi}{4}\right)=\frac{1}{2}=-\frac{1}{2}+C$$

$C = 1$

$$y(x)=\frac{-1}{1+\tan x}+1$$

$$y(x)=\frac{-1+1+\tan x}{1+\tan x}$$

$$y(x)=\frac{\tan x}{1+\tan x}$$

Solving with $y=\sqrt{2}\sin x$

$$\frac{\tan x}{1+\tan x}=\sqrt{2}\sin x$$

$\sin x=0,$ $\qquad \frac{1}{\sqrt{2}}=\sin x+\cos x$

$x=\pi$ $\qquad \frac{1}{2}=\sin\left(x+\frac{\pi}{4}\right)$

$$\sin\frac{\pi}{6}=\sin\left(x+\frac{\pi}{4}\right)$$

$$x+\frac{\pi}{4}=\pi-\frac{\pi}{6},2\pi+\frac{\pi}{6}$$

$$x=\frac{5\pi}{6}-\frac{\pi}{4},x=\frac{13\pi}{6}-\frac{\pi}{4}$$

$$x=\frac{7\pi}{12},x=\frac{23\pi}{12}$$

sum of sol.

$$=\pi+\frac{7\pi}{12}+\frac{23\pi}{12}$$

$$=\frac{12\pi+7\pi+23}{12}=\frac{42\pi}{12}=\frac{k\pi}{12}$$

$\Rightarrow k=42$

45. Let $y=y(x)$ be the solution of the differential equation $\left(1-x^2\right)dy=\left(xy+\left(x^3+2\right)\sqrt{1-x^2}\right)dx,-1<x<1$ and $y(0)=0$. If $\int_{-\frac{1}{2}}^{\frac{1}{2}}\sqrt{1-x^2}\,y(x)\,dx=k$, then k^{-1} is equal to _____. **[JEE(Main)June2022]**

माना अवकल समीकरण $\left(1-x^2\right)dy=\left(xy+\left(x^3+2\right)\sqrt{1-x^2}\right)dx,-1<x<1$, $y(0)=0$, का हल $y=y(x)$ है। यदि $\int_{-\frac{1}{2}}^{\frac{1}{2}}\sqrt{1-x^2}\,y(x)\,dx=k$ है, तो k^{-1} बराबर है _____ **[JEE(Main)June2022]**

Ans. 320

Sol. $\frac{dy}{dx}=\frac{xy}{1-x^2}+\frac{x^3+2}{\sqrt{1-x^2}}$

$\frac{dy}{dx}+\left(\frac{-x}{1-x^2}\right)y=\frac{x^3+2}{\sqrt{1-x^2}}$

I.F. $= e^{\int\frac{-x}{1-x^2}dx}=\sqrt{1-x^2}$

$y\left(\sqrt{1-x^2}\right)=\int\left(x^3+2\right)dx+c$

$\Rightarrow y\sqrt{1-x^2}=\frac{x^4}{4}+2x+c$

$x = 0, y = 0$

$0 = 0 + 0 + c \qquad \Rightarrow c = 0$

$\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\frac{x^4}{4}+2x\right)dx$

$\left.\frac{x^5}{20}+x^2\right|_{-\frac{1}{2}}^{\frac{1}{2}}=\frac{1}{20}\left(\frac{1}{32}+\frac{1}{32}\right)=\frac{1}{320}$

$k^{-1} = 320$

46. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx}+\frac{\sqrt{2}y}{2\cos^4 x-\cos 2x}=xe^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$, $0<x<\frac{\pi}{2}$ with $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{32}$. If $y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to _____.

[JEE(Main)June2022]

माना अवकल समीकरण $\frac{dy}{dx}+\frac{\sqrt{2}y}{2\cos^4 x-\cos 2x}=xe^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$, $0<x<\frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{32}$ का हल $y = y(x)$ है। यदि $y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$ है, तो $3\alpha^2$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 2

Sol. $\frac{dy}{dx}+\frac{\sqrt{2}y}{2\cos^4 x-\cos 2x}=xe^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$, $0<x<\frac{\pi}{2}$

$\frac{dy}{dx}+\frac{2\sqrt{2}y}{1+\cos^2 2x}=xe^{\tan^{-1}\left(\sqrt{2}\cot+2x\right)}$ .....(1)

$$\text{I.F.} = e^{\int \frac{2\sqrt{2}}{1+\cos^2 2x}dx} = e^{\int \frac{2\sqrt{2}\sec^2 2x}{\tan^2 2x+2}dx} = e^{\tan^{-1}\left(\frac{\tan 2x}{\sqrt{2}}\right)}$$

Solution of (1)

$$y.e^{\tan^{-1}\left(\frac{\tan^2 x}{\sqrt{2}}\right)} = \int x.e^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)+\tan^{-1}\left(\frac{1}{\sqrt{2}\cot 2x}\right)}dx$$

$$y.e^{\tan^{-1}\left(\frac{\tan^2 x}{\sqrt{2}}\right)} = e^{\frac{\pi}{2}}\frac{x^2}{2} + c$$

$\because \; y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} \Rightarrow c = 0$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}\left(\frac{\tan\frac{2\pi}{3}}{\sqrt{2}}\right)} = \frac{\pi}{18}e^{-\tan^{-1}(\alpha)}$$

$$\Rightarrow \frac{\pi}{2} - \tan^{-1}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right) = -\tan^{-1}(\alpha)$$

$$\Rightarrow \alpha = +\sqrt{\frac{2}{3}} \Leftarrow 3\alpha^2 = 2 \text{ Ans.}$$

47. Let $y = y(x)$, $x > 1$, be the solution of the differential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha + 1}{\beta e^\alpha}$, then the value of $\alpha + \beta$ is equal to _____. **[JEE(Main)June2022]**

माना अवकल समीकरण $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, $y(2) = \frac{1+e^4}{2e^4}$ का हल $y = y(x)$, $x > 1$ है। यदि $y(3) = \frac{e^\alpha + 1}{\beta e^\alpha}$ है, तो $\alpha + \beta$ का मान बराबर है _____. **[JEE(Main)June2022]**

Ans. 14

Sol. $$\frac{dy}{dx} + \frac{2x}{(x-1)}.y = \frac{1}{(x-1)^2}$$

$$\text{I.F.} = e^{\int p dx} = e^{\int \frac{2x}{(x-1)}dx} = e^{\int\left(2+\frac{2}{x-1}\right)dx}$$

$$= e^{2x+2\log(x-1)}$$

$$= e^{2x}.e^{\log(x-1)^2}$$

$= (x-1)^2 . e^{2x}$

$\therefore\ y.(x-1)^2 e^{2x} = \int \frac{1}{(x-1)^2}.(x-1)^2 e^{2x}.dx + c$

$\therefore\ y.(x-1)^2 e^{2x} = \frac{e^{2x}}{2} + c$

$\because\ y(2) = \frac{1+e^4}{2e^4}$

$\frac{(1+e^4)}{2e^4}.1.e^4 = \frac{e^4}{2} + c$

$c = \frac{1}{2}$

$y(x-1)^2 e^{2x} = \frac{e^{2x}}{2} + \frac{1}{2}$

If $y(3) = \frac{e^\alpha + 1}{\beta.e^\alpha}$

$\frac{e^\alpha + 1}{\beta.e^\alpha}.4.e^6 = \frac{e^6}{2} + \frac{1}{2}$

$\frac{e^\alpha + 1}{\beta.e^\alpha} = \frac{e^6 + 1}{8.e^6}$

$\alpha = 6, \beta = 8$

$\alpha + \beta = 14$

48. The general solution of the differential equation $(x - y^2)dx + y(5x + y^2)dy = 0$ is :

[JEE(Main)July2022]

अवकल समीकरण $(x - y^2)dx + y(5x + y^2)dy = 0$ का व्यापक हल है : **[JEE(Main)July2022]**

(A) $(y^2 + x)^4 = C|(y^2 + 2x)^3|$ (B) $(y^2 + 2x)^4 = C|(y^2 + x)^3|$

(C) $|(y^2 + x)^3| = C(2y^2 + x)^4$ (D) $|(y^2 + 2x)^3| = C(2y^2 + x)^4$

Ans. A

Sol. $(x - y^2)dx + y(5x + y^2)dy = 0$

$y\frac{dy}{dx} = \frac{y^2 - x}{5x + y^2}$

Let $y^2 = t$

$$\frac{1}{2}\cdot\frac{dt}{dx} = \frac{t-x}{5x+t}$$

Now substitute, $t = vx$

$$\frac{dt}{dx} = v + x\frac{dv}{dx}$$

$$\frac{1}{2}\left\{v + x\frac{dv}{dx}\right\} = \frac{v-1}{5+v}$$

$$x\frac{dv}{dx} = \frac{2v-2}{5+v} - v = \frac{-3v-v^2-2}{5+v}$$

$$\int\frac{5+v}{v^2+3v+2}dv = \int -\frac{dx}{x}$$

$$\int\frac{4}{v+1}dv - \int -\frac{3}{v+2}dv = -\int\frac{dx}{x}$$

$4\ln|v+1| - 3\ln|v+2| = -\ln x + \ln C$

$$\left|\frac{(v+1)^4}{(v+2)^3}\right| = \frac{C}{x}$$

$$\left|\frac{\left(\frac{y^2}{x}+1\right)^4}{\left(\frac{y^2}{x}+2\right)^3}\right| = \frac{C}{x}$$

$$\left|\left(y^2+x\right)^4\right| = C\left|\left(y^2+2x\right)^3\right|$$

49. Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points (1, 2) and (8, 1), then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to :

[JEE(Main)July2022]

माना एक निषकोण वक्र $y = f(x)$ के किसी भी बिन्दु (x, y) पर स्पर्श रेखा की प्रवणता $\left(\frac{-y}{x}\right)$ के अनुक्रमानुपात में है। यदि यह वक्र बिन्दुओं (1, 2) तथा (8, 1) से होकर जाता है, तो $\left|y\left(\frac{1}{8}\right)\right|$ बराबर है : **[JEE(Main)July2022]**

(A) $2\log_e 2$ (B) 4 (C) 1 (D) $4\log_e 2$

Ans. B

Sol. $\frac{dy}{dx} \propto \frac{-y}{x}$

$\frac{dy}{dx} = \frac{-ky}{x} \Rightarrow \int \frac{dy}{y} = -K\int \frac{dx}{x}$

$\ln|y| = -K\ln|x| + C$

If the above equation staisfy (1, 2) and (8, 1)

$\ln 2 = -K \times 0 + C \Rightarrow C = \ln 2$

$\ln 1 = -K\ln 8 + \ln 2 \Rightarrow K = \frac{1}{3}$

So, at $x = \frac{1}{8}$

$\ln|y| = -\frac{1}{3}\ln\left(\frac{1}{8}\right) + \ln 2 = 2\ln 2$

$|y| = 4$

50. If $\frac{dy}{dx} + 2y\tan x = \sin x, 0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of y(x) is :

[JEE(Main)July2022]

यदि $\frac{dy}{dx} + 2y\tan x = \sin x, 0 < x < \frac{\pi}{2}$, $y\left(\frac{\pi}{3}\right) = 0$ हैं, तो y(x) का अधिकतम मान है : **[JEE(Main)July2022]**

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Ans. A

Sol. $\frac{dy}{dx} + 2y\tan x = \sin x$

which is a first order linear differential equation.

Integrating factor (I.F.) $= e^{\int 2\tan x dx}$

$= e^{2\ln|\sec x|} = \sec^2 x$

Solution of differential equation can be written as

$y.\sec^2 x = \int \sin x.\sec^2 x dx = \int \sec x.\tan x dx$

$y\sec^2 x = \sec x + C$

$$y\left(\frac{\pi}{3}\right)=0,0=\sec\frac{\pi}{3}+C\Rightarrow\ C=-2$$

$$y=\frac{\sec x-2}{\sec^2 x}=\cos x-2\cos^2 x$$

$$=\frac{1}{8}-2\left(\cos x-\frac{1}{4}\right)^2$$

$$y_{max}=\frac{1}{8}$$

51. Let the solution curve y = f(x) of the differential equation $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$, $x\in(-1,1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ is equal to : **[JEE(Main)July2022]**

माना अवकल समीकरण $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$, $x\in(-1,1)$ का हल वक्र $y=f(x)$ मूल बिन्दु से होकर जाता है। तो $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ बराबर है : **[JEE(Main)July2022]**

(A) $\frac{\pi}{3}-\frac{1}{4}$ (B) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6}-\frac{\sqrt{3}}{2}$

Ans. B

Sol. $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$

$$I.F=e^{\int\frac{x}{x^2-1}dx}$$

$$I.F=\sqrt{1-x^2}$$

Solution of D.E.

$$y.\sqrt{1-x^2}=\int\frac{x^4+2x}{\sqrt{1-x^2}}\sqrt{1-x^2}\,dx$$

$$y.\sqrt{1-x^2}=\int(x^4+2x)\,dx$$

$$y.\sqrt{1-x^2} = \frac{x^5}{5} + x^2 + C$$

At $x = 0, y = 0$, get $C = 0$

$$y = \frac{x^5}{5\sqrt{1-x^2}} + \frac{x^2}{\sqrt{1-x^2}}$$

Now,

$$\int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx = \int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^5}{5\sqrt{1-x^2}}\,dx + \int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx$$

$$\int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx = 0 + 2\int_{0}^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx$$

$$\int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$$

52. Let $y = y_1(x)$ and $y = y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx} = x + y$, with $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then, the number of points of intersection of $y = y_1(x)$ and $y = y_2(x)$ is : **[JEE(Main)July2022]**

माना अवकल सीमकरण $\frac{dy}{dx} = x + y$ के $y_1(0) = 0$ तथा $y_2(0) = 1$ के लिए दो भिन्न हल क्रमशः $y = y_1(x)$ तथा $y = y_2(x)$ के प्रतिच्छेदन बिन्दुओं की संख्या है : **[JEE(Main)July2022]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\frac{dy}{dx} = x + y \Rightarrow \frac{dy}{dx} - y = x$

$1f = e^{-x}$

$\therefore$ solution is $ye^{-x} = \int xe^{-x}dx$

$\Rightarrow ye^{-x} = -xe^{-x} - e^{-x} + c$

$\Rightarrow y = -x - 1 + ce^{x}$

$y_1(0) = 0 \Rightarrow c = 1$

$\therefore y_1 = -x - 1 + e^x$ .....(1)

$y_2(0) = 1 \Rightarrow c = 2$

$\therefore\ y_2 = -x - 1 + 2e^x$ .....(2)

Now $y_2 - y_1 = e^x > 0 \therefore y_2 \neq y_1$

$\therefore$ Number of points of intersection of y_1 & y_2 is zero.

53. Let the solution curve of the differential equation $xdy = \left(\sqrt{x^2+y^2}+y\right)dx$, $x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is : **[JEE(Main)July2022]**

माना अवकल समीकरण $xdy = \left(\sqrt{x^2+y^2}+y\right)dx$, $x > 0$ का हल वक्र, रेखा $x = 1$ को $y = 0$ पर तथा रेखा $x = 2$ को $y = \alpha$ पर काटता है। तो α का मान है : **[JEE(Main)July2022]**

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. B

Sol. $xdy = \left(\sqrt{x^2+y^2}+y\right)dx$

$xdy - ydx = \sqrt{x^2+y^2}dx$

$\frac{xdy-ydx}{x^2} = \sqrt{1+\frac{y^2}{x^2}}.\frac{dx}{x}$

$\frac{d(y/x)}{\sqrt{1+\left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$

$\ln\left(\frac{y}{x}+\sqrt{\left(\frac{y}{x}\right)^2+1}\right) = \ln x + R$

$\frac{y+\sqrt{y^2+x^2}}{x} = cx$

$y+\sqrt{y^2+x^2} = cx^2$

$x = 1, y = 0 \Rightarrow 0 + 1 = C \Rightarrow C = 1$

Curve is $y+\sqrt{x^2+y^2} = x^2$

$x = 2, y = \alpha$

$2+\sqrt{4+\alpha^2} = 4$

$4+\alpha^2 = 16+\alpha^2 = 8\alpha$

$$\alpha = \frac{3}{2}$$

54. If $y = y(x)$, $x \in (0, \pi/2)$ be the solution curve of the differential equation

$\left(\sin^2 2x\right)\frac{dy}{dx} + \left(8\sin^2 2x + 2\sin 4x\right)y = 2e^{-4x}\left(2\sin 2x + \cos 2x\right)$, with $y\left(\frac{\pi}{4}\right) = e^{-\pi}$, then $y\left(\frac{\pi}{6}\right)$ is equal to :

[JEE(Main)July2022]

यदि अवकल समीकरण $\left(\sin^2 2x\right)\frac{dy}{dx} + \left(8\sin^2 2x + 2\sin 4x\right)y = 2e^{-4x}\left(2\sin 2x + \cos 2x\right)$, $x \in (0, \pi/2)$

$y\left(\frac{\pi}{4}\right) = e^{-\pi}$ का हल वक्र $y = y(x)$ है, तो $y\left(\frac{\pi}{6}\right)$ बराबर है : **[JEE(Main)July2022]**

(A) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}e^{2\pi/3}$ (C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Ans. A

Sol. Given differential equation can be re-written as

$$\frac{dy}{dx} + \left(8 + 4\cot 2x\right)y = \frac{2e^{-4x}}{\sin^2 2x}\left(2\sin x + \cos 2x\right)$$

which is a linear diff. equation.

$$\text{I.f.} = e^{\int(8+4\cot 2x)dx} = e^{8x+2Cu(\sin 2x)}$$

$$= e^{8x}.\sin^2 2x$$

$\therefore$ solution is

$$y\left(e^{8x}.\sin^2 2x\right) = \int 2e^{4x}\left(2\sin 2x + \cos 2x\right)dx + C$$

$$= e^{4x}.\sin 2x + C$$

Given $y\left(\frac{\pi}{4}\right) = e^{-\pi} \Rightarrow C = 0$

$$\therefore y = \frac{e^{-4x}}{\sin 2x}$$

$$\therefore y\left(\frac{\pi}{6}\right) = \frac{e^{-4.\frac{\pi}{6}}}{\sin\left(2.\frac{\pi}{6}\right)} = \frac{2}{\sqrt{3}}e^{-\frac{2\pi}{3}}$$

55. The minimum value of the twice differentiable function $f(x)=\int_0^x e^{x-t}f'(t)dt-\left(x^2-x+1\right)e^x, x\in R$, is :

[JEE(Main)July2022]

दो बार अवकलनीय फलन $f(x)=\int_0^x e^{x-t}f'(t)dt-\left(x^2-x+1\right)e^x, x\in R$ का निम्नतम मान है :

[JEE(Main)July2022]

(A) $-\frac{2}{\sqrt{e}}$ (B) $-2\sqrt{e}$ (C) $-\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Ans. A

Sol. $f(x)=e^x.\int_0^x\frac{f'(t)}{e^t}dt$

$f'(x)=e^x.\int_0^x\frac{f'(t)}{e^t}dt+e^x.\frac{f'(x)}{e^x}$

$-\left[(2x-1).e^x+\left(x^2-x+1\right).e^x\right]$

$\int_0^x\frac{f'(t)}{e^t}dt=x^2+x$

$\frac{f'(x)}{e^x}=2x+1$

$f'(x)=(2x+1).e^x$

$f'(x)=0\Rightarrow x=-\frac{1}{2}$

$f(x)=(2x+1).e^x-2e^x+C$

$|f(0)=-1$

$-1 = 1 - 2 + C$

$C = 0$

$f(x) = e^x (2x - 1)$

$f\left(-\frac{1}{2}\right)=\frac{-2}{\sqrt{e}}$

56. Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$, $x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}\ y(8)$ is : **[JEE(Main)July2022]**

माना अवकल समीकरण $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$, $x > 1$ का हल वक्र $y = y(x)$ बिन्दु $\left(2, \sqrt{\frac{1}{3}}\right)$ से होकर जाता है। तो $\sqrt{7}$ $y(8)$ होगा : **[JEE(Main)July2022]**

(A) $11 + 6\log_e 3$ (B) 19 (C) $12 - 2\log_e 3$ (D) $19 - 6\log_e 3$

Ans. D

Sol. $\frac{dy}{dx} + \frac{1}{x^2-1}y\left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$,

$\frac{dy}{dx} + Py = Q$

$I.F. = e^{\int P dx} = \left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$

$y\left(\frac{x-1}{x+1}\right)^{\frac{1}{2}} = \int\left(\frac{x-1}{x+1}\right)^{1} dx$

$= x - 2\log_e|x+1| + C$

Curve passes through $\left(2, \frac{1}{\sqrt{3}}\right)$

$\Rightarrow C = 2\log_e 3 - \frac{5}{3}$

at $x = 8$,

$\sqrt{7}y(8) = 19 - 6\log_e 3$

57. The differential equation of the family of circles passing through the points (0, 2) and (0, –2) is : **[JEE(Main)July2022]**

बिन्दुओं (0, 2) तथा (0, –2) से होकर जाने वाले वृत्त कुल का अवकल समीकरण है : **[JEE(Main)July2022]**

(A) $2xy\frac{dy}{dx} + \left(x^2 - y^2 + 4\right) = 0$ (B) $2xy\frac{dy}{dx} + \left(x^2 + y^2 - 4\right) = 0$

(C) $2xy\frac{dy}{dx}+\left(y^2-x^2+4\right)=0$ (D) $2xy\frac{dy}{dx}-\left(x^2-y^2+4\right)=0$

Ans. A

Sol. Equation of circle passing through (0, –2) and (0, 2) is

$x^2+\left(y^2-4\right)+\lambda x=0,(\lambda\in R)$

Divided by x we get

$\frac{x^2+\left(y^2-4\right)}{x}+\lambda=0$

Differentiating with respect to x

$\frac{x\left[2x+2y.\frac{dy}{dx}\right]-\left[x^2+y^2-4\right].1}{x^2}=0$

$\Rightarrow 2xy.\frac{dy}{dx}+\left(x^2-y^2+4\right)=0$

58. Let the solution curve y = y(x) of the differential equation $\left(1+e^{2x}\right)\left(\frac{dy}{dx}+y\right)=1$ pass through the point $\left(0,\frac{\pi}{2}\right)$.

Then, $\lim_{x\to\infty}e^x y(x)$ is equal to : **[JEE(Main)July2022]**

माना अवकल समीकरण $\left(1+e^{2x}\right)\left(\frac{dy}{dx}+y\right)=1$ का हल वक्र y = y(x), $\left(0,\frac{\pi}{2}\right)$ से होकर जाता है। तो $\lim_{x\to\infty}e^x y(x)$ बराबर है : **[JEE(Main)July2022]**

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

Ans. B

Sol. $\frac{dy}{dx}+y=\frac{1}{1+e^{2x}}$

So integrating factor is $e^{\int 1.dx}=e^x$

So solution is $y.e^x=\tan^{-1}\left(e^x\right)+c$

Now as curve is passing through $\left(0,\frac{\pi}{2}\right)$ so

$\Rightarrow c=\frac{\pi}{4}$

$$\Rightarrow \lim_{x\to\infty}\left(y.e^x\right)=\lim_{x\to\infty}\left(\tan^{-1}\left(e^x\right)+\frac{\pi}{4}\right)=\frac{3\pi}{4}$$

59. If the solution curve of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passes through the points (2, 1) and (k + 1, 2), k > 0, then : **[JEE(Main)July2022]**

यदि अवकल समीकरण $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ का हल वक्र बिन्दुओं (2, 1) तथा (k + 1, 2), k > 0, से होकर जाता है, तो : **[JEE(Main)July2022]**

(A) $2\tan^{-1}\left(\frac{1}{k}\right)=\log_e\left(k^2+1\right)$ (B) $\tan^{-1}\left(\frac{1}{k}\right)=\log_e\left(k^2+1\right)$

(C) $2\tan^{-1}\left(\frac{1}{k+1}\right)=\log_e\left(k^2+2k+2\right)$ (D) $2\tan^{-1}\left(\frac{1}{k}\right)=\log_e\left(\frac{k^2+1}{k^2}\right)$

Ans. A

Sol. $\frac{dy}{dx}=\frac{x+y-2}{x-y}=\frac{(x-1)+(y-1)}{(x-1)-(y-1)}$

$x-1=X, y-1=Y$

$\frac{dY}{dX}=\frac{X+Y}{X-Y}$

$Y=VX \quad \frac{dY}{dX}=V+X\frac{dV}{dX}$

$V+X\frac{dV}{dX}=\frac{1+V}{1-V} \qquad x\frac{dV}{dX}=\frac{v^2+1}{1-v}$

$\int\frac{1-V}{1+V^2}dV=\int\frac{dX}{X}$

$\int\frac{dV}{1+V^2}-\frac{1}{2}\int\frac{2VdV}{1+V^2}=\int\frac{dX}{XX}$

$\tan^{-1}V-\frac{1}{2}\ln\left(1+V^2\right)=\ln X+c$

$\tan^{-1}\left(\frac{Y}{X}\right)-\frac{1}{2}\ln\left(1+\frac{Y^2}{X^2}\right)=\ln(X)+c$

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\frac{(y-1)^2}{(x-1)^2}\right)=\ln(x-1)+c$

Passes through (2, 1)

$$\therefore \tan^{-1}\left(\frac{1}{k}\right)-\frac{1}{2}\ln\left(1+\frac{1}{k^2}\right)=\ln k$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\ln\left(\frac{1+k^2}{k^2}\right)+2\ln k$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\ln\left(1+k^2\right)$$

60. Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{(x+3)}{x+1}$, $x>-1$, which passes through the point (0, 1). Then y(1) is equal to : **[JEE(Main)July2022]**

माना $\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{(x+3)}{x+1}$, $x<-1$ का हल वक्र y = y(x), बिन्दु (0, 1) से होकर जाता है। तो y(1) बराबर है : **[JEE(Main)July2022]**

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Ans. B

Sol. $\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{x+3}{x+1}$

$\int p(x)dx$ I.F. $=e^{\int p(x)dx}$

$$\int p(x)dx=\int\frac{\left(2x^2+11x+13\right)dx}{(x+1)(x+2)(x+3)}$$

Using partial fraction

$$\frac{2x^2+11x+13}{(x+1)(x+2)(x+3)}=\frac{A}{x+1}+\frac{B}{x+2}+\frac{C}{x+3}$$

$A=\frac{4}{2}=2$

$B=1$

$C=-1$

$\because \int p(x)dx=A\ln(x+1)+B\ln(x+2)+c\ln(x+3)$

$$=\ln\left(\frac{(x+1)^2(x+2)}{x+3}\right)$$

I.F. $= e^{\int p(x)dx} = \dfrac{(x+1)^2(x+2)}{(x+3)}$

Solution $y(IF) = \int Q \cdot (IF)dx$

$$y\left(\frac{(x+1)^2(x+2)}{x+3}\right) = \int\left(\frac{x+3}{x+1}\right)\frac{(x+1)\cancel{(x+2)}}{(x+3)}dx$$

$$y\left(\frac{(x+1)^2(x+2)}{x+3}\right) = \frac{x^3}{3} + \frac{3x^2}{2} + 2x + c$$

Passes through (0, 1) $C = \dfrac{2}{3}$

$$\Rightarrow y(1) = \frac{3}{2}$$

61. Let y = y(x) be the solution of the differential equation $\dfrac{dy}{dx} = \dfrac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1$. If for some $n \in N$, $y(2) \in [n-1, n)$, then n is equal to _____. **[JEE(Main)July2022]**

माना अवकल समीकरण $\dfrac{dy}{dx} = \dfrac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1$ का हल y = y(x) है। यदि किसी $n \in N$ के लिए $y(2) \in [n-1, n)$ है, तो n बराबर है _____ **[JEE(Main)July2022]**

Ans. 3

Sol. $$\frac{dy}{dx} = \frac{y}{x}\frac{\left(4y^2 + 2x^2\right)}{\left(3y^2 + x^2\right)}$$

Put $y = vx$ $\Rightarrow$ $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{v\left(4v^2 + 2\right)}{\left(3v^2 + 1\right)}$$

$$\Rightarrow x\frac{dv}{dx} = v\left(\frac{\left(4v^2 + 2 - 3v^2 - 1\right)}{3v^2 + 1}\right)$$

$$\int\left(3v^2 + 1\right)\frac{dv}{v^3 + v} = \int\frac{dx}{x}$$

$\Rightarrow \ln|v^3 + v| = \ln x + c$

$\Rightarrow \left|\left(\frac{y}{x}\right)^3 + \left(\frac{y}{x}\right)\right| = \ln x + C$

$\downarrow y(1) = 1$

$\Rightarrow C = \ln 2$

$\therefore$ for y(2)

$$\ln\left(\frac{y^3}{8} + \frac{y}{2}\right) = 2\ln 2 \Rightarrow \frac{y^3}{8} + \frac{y}{2} = 4$$

$\Rightarrow [y(2)] = 2$

$\Rightarrow n = 3$

62. Let a curve y = y(x) pass through the point (3, 3) and the area of the region under this curve, above the x-axis and between the abscissae 3 and x(> 3) be $\left(\frac{y}{X}\right)^3$. If this curve also passes through the point $\left(\alpha, 6\sqrt{10}\right)$ in the first quadrant, then α is equal to _____. **[JEE(Main)July2022]**

माना एक वक्र y = y(x) बिन्दु (3, 3) से होकर जाता है तथा इस वक्र के नीचे, x-अक्ष के ऊपर तथा भुजों 3 और x(> 3) के बीच के क्षेत्र का क्षेत्रफल $\left(\frac{y}{X}\right)^3$ है। यदि वह वक्र प्रथम चतुर्थांश में बिन्दु $\left(\alpha, 6\sqrt{10}\right)$ से भी होकर जाता है, तो α बराबर है _____

[JEE(Main)July2022]

Ans. 6

Sol. $x^4 = 3yx.y' - 3y^2$

$$\Rightarrow 3xy\frac{dy}{dx} = 3y^2 + x^4$$

Put $y^2 = t, y\frac{dy}{dx} = \frac{1}{2}\frac{dt}{dx}$

$$\frac{dt}{dx} - \frac{2}{x}t = \frac{2}{3}x^3$$

$$\therefore \frac{t}{x^2} = \frac{x^2}{3} + C$$

$$\Rightarrow \frac{y^2}{x^2} = \frac{x^2}{3} - 2$$

Put (3, 3) , C = –2

$\therefore \frac{y^2}{x^2} = \frac{x^2}{3} - 2$

$3y^2 = x^4 - 6x^2$

$x^4 - 6x^2 = 1080$

$\therefore x = 6$

63. Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim\limits_{x\to\infty} y(x)$ is finite. If a and b are respectively the x- and y- intercepts of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _____. **[JEE(Main)July2022]**

माना अवकल समीकरण $\frac{dy}{dx} - y = 2 - e^{-x}$ के हल वक्र $\lim\limits_{x\to\infty} y(x)$ परिमित है। यदि वक्र की $x = 0$ पर स्पर्श रेखा के x – तथा y – अंतः खण्ड क्रमशः a तथा b हैं, तो $a - 4b$ का मान बराबर है _____ **[JEE(Main)July2022]**

Ans. 3

Sol. $\frac{dy}{dx} - y = 2 - e^{-x}$

I.F. $= e^{-\int dx} = e^{-x}$

$\therefore$ solution of D.E.

$y.e^{-x} = \int \left(2e^{-x} - e^{-2x}\right) dx$

$\Rightarrow y = -2 + \frac{e^{-x}}{2} + C.e^{x}$

$\because \lim\limits_{x\to\infty} y$ is finite

$\because \lim\limits_{x\to\infty} \left(-2 + \frac{e^{-x}}{2} + C.e^{x}\right) \to$ finite

This is possible only when $C = 0$

$\therefore y = y(x) = -2 + \frac{e^{-x}}{2}$

$\frac{dy}{dx} = -\frac{1}{2}e^{-x}$

$\left.\frac{dy}{dx}\right|_{x=0} = -\frac{1}{2} = m, y(0) = -2 + \frac{1}{2} = \frac{-3}{2}$

$\therefore$ equation of tangent

$$y+\frac{3}{2}=-\frac{1}{2}(x-0)$$

$\Rightarrow x + 2y = -3$

$$a=-3, b=\frac{-3}{2}$$

$a - 4b = -3 + 6 = 3$

64. Let $y = y(x)$ be the solution curve of the differential equation

$\sin\left(2x^2\right)\log_e\left(\tan x^2\right)dy+\left(4xy-4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)\right)dx=0$, $0 < x \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}},1\right)$. then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to _____. **[JEE(Main)July2022]**

माना अवकल समीकरण $\sin\left(2x^2\right)\log_e\left(\tan x^2\right)dy+\left(4xy-4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)\right)dx=0$, $0 < x \sqrt{\frac{\pi}{2}}$ का हल वक्र $y = y(x)$, बिन्दु $\left(\sqrt{\frac{\pi}{6}},1\right)$ से होकर जाता है। तो $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ बराबर है _____. **[JEE(Main)July2022]**

Ans. 1

Sol. $\sin\left(2x^2\right)\ln\left(\tan x^2\right)dy+\left(4xy-\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)\right)dx=0$

$$\ln\left(\tan x^2\right)dy+\frac{4xydx}{\sin\left(2x^2\right)}-\frac{4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)}{\sin\left(2x^2\right)}dx=0$$

$$d\left(y.\ln\left(\tan x^2\right)\right)-4\sqrt{2}x\frac{\left(\sin x^2-\cos x^2\right)}{\sqrt{2}-2\sin x^2\cos x^2}dx=0$$

$$d\left(y\ln\left(\tan x^2\right)\right)-\frac{4x\left(\sin x^2-\cos x^2\right)}{\left(\sin x^2+\cos^2\right)-1}dx=0$$

$$\Rightarrow \int d\left(y\ln\left(\tan x^2\right)\right)+2\int\frac{dt}{t^2-1}=\int 0$$

$$\Rightarrow y\ln\left(\tan x^2\right)+2.\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|=c$$

$$y\ln\left(\tan x^2\right)+\ln\left(\frac{\sin x^2+\cos x^2-1}{\sin x^2+\cos x^2+1}\right)=c$$

Put $y=1$ and $x=\sqrt{\frac{\pi}{6}}$

$$1\ln\left(\frac{1}{\sqrt{3}}\right)+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}=c$$

$$\text{Now } x=\sqrt{\frac{\pi}{3}}\Rightarrow y\left(\ln\sqrt{3}\right)+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}=\ln\left(\frac{1}{\sqrt{3}}\right)+o\ln\left(\frac{\sqrt{3}-1}{\sqrt{3}+3}\right)$$

$$y\left(\ln\sqrt{3}\right)=\ln\left(\frac{1}{\sqrt{3}}\right)$$

$\Rightarrow y=-1$

$|y|=1$